

Signals and Systems

Previous Year Questions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Signals and Systems (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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PrepFusion

UNIT

1

Basics of Signals

Topics

1. Basics of Signals and Mathematical Foundation
2. Step and Rectangular Signal
3. Ramp and Triangular Signal
4. Impulse Signal
5. Miscellaneous Signals
6. Complex Exponential Signal
7. Periodicity of a Signal
8. Orthogonality of Signals
9. Energy, Average and Power of a Signal
10. Even, Odd, Conjugate Symmetry Signals
11. Discrete-Time Signals and Their Types
12. Classification of Discrete-Time Signals

Unit Snapshot*

Stream: EC	**Questions:** 15	**MCQ:** 10	**MSQ:** 3	**NAT:** 2
1M: 11	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2015)–4M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M

Consider a real signal $x(t)$, $-\infty < t < \infty$, such that $x(t) = 0$ for $t < 0$, $x(t) = 2$ for $0 \le t < 1$ and $x(t) = 0$ for $t \ge 1$.
Let $E[x(t)] = \int_{-\infty}^{\infty} [x(t)]^2\, dt$.
Which of the following options correctly represents the ratio, $E[x(t)]/E[3\,x(-3t+5)]$?

A. 3 **B.** 1 **C.** $1/3$ **D.** $1/9$

Q1.26.2 MSQ 2026 2M

Let $x_1(t) = \cos(2\pi nt)$ and $x_2(t) = 2\sin(4\pi nt)$ represent two sinusoids for a positive integer n and $-\infty < t < \infty$.
Which of the following statements about $x_1(t)$ and $x_2(t)$ is/are valid?

- ☐ **A.** $x_1(t)$ and $x_2(t)$ are orthogonal to each other over $0 \le t < 1/n$.
- ☐ **B.** $x_1(t)$ and $x_2(t)$ are orthonormal to each other over $0 \le t < 1/n$.
- ☐ **C.** $x_2(t)$ is a harmonic of $x_1(t)$.
- ☐ **D.** $x_1(t)$ and $x_2(t)$ are non-orthogonal to each other over $0 \le t < 1/(2n)$.

Q1.24.1 MSQ 2024 2M

The radian frequency value(s) for which the discrete time sinusoidal signal $x[n] = A\cos(\Omega n + \pi/3)$ has a period of 40 is/are

- ☐ **A.** 0.15π
- ☐ **B.** 0.225π
- ☐ **C.** 0.3π
- ☐ **D.** 0.45π

Q1.19.1 NAT 2019 1M

Consider the signal $f(t) = 1 + 2\cos(\pi t) + 3\sin\left(\frac{2\pi}{3}t\right) + 4\cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$, where t is in seconds. Its fundamental time period, in seconds, is _____.

Q1.15.1 NAT 2015 1M

The waveform of a periodic signal $x(t)$ is shown in the figure.

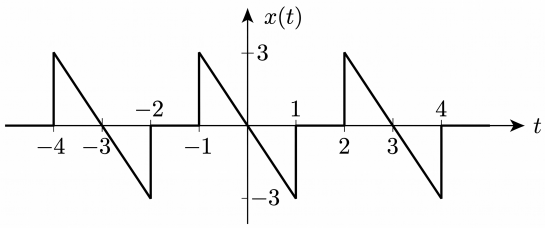


Fig. Q1.15.1

A signal $g(t)$ is defined by $g(t) = x\left(\frac{t-1}{2}\right)$. The average power of $g(t)$ is _____.

Q1.14.1 MCQ 2014 1M

A discrete-time signal $x[n] = \sin(\pi^2 n)$, n being an integer, is

A. Periodic with period π.

B. Periodic with period π^2.

C. Periodic with period $\pi/2$.

D. Not periodic.

Q1.13.1 MCQ 2013 1M

For a periodic signal $v(t) = 30\sin 100t + 10\cos 300t + 6\sin(500t + \pi/4)$, the fundamental frequency in rad/s is

A. 100 **B.** 300 **C.** 500 **D.** 1500

Q1.09.1 MCQ 2009 1M

A function is given by $f(t) = \sin^2 t + \cos 2t$. Which of the following is true ?

A. f has frequency components at 0 and $1/2\pi$ Hz

B. f has frequency components at 0 and $1/\pi$ Hz

C. f has frequency components at $1/2\pi$ and $1/\pi$ Hz

D. f has frequency components at 0, $1/2\pi$ and $1/\pi$ Hz

Q1.06.1 MCQ 2006 1M

The Dirac delta function $\delta(t)$ is defined as

A. $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{otherwise} \end{cases}$

B. $\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & \text{otherwise} \end{cases}$

C. $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{otherwise} \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t)dt = 1$

D. $\delta(t) = \begin{cases} \infty & t = 0 \\ 0 & \text{otherwise} \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t)dt = 1$

Q1.05.1 MCQ 2005 1M

The power in the signal $s(t) = 8\cos\left(20\pi t - \frac{\pi}{2}\right) + 4\sin(15\pi t)$ is:

A. 40 **B.** 41 **C.** 42 **D.** 82

Q1.05.2 MCQ 2005 1M

The function $x(t)$ is shown in figure. Even and odd parts of a unit-step function $u(t)$ are respectively.

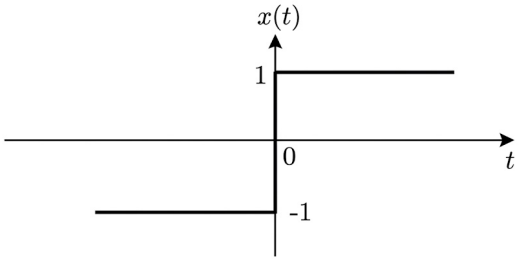


Fig. Q1.05.2

A. $\frac{1}{2}, \frac{1}{2}x(t)$ **B.** $-\frac{1}{2}, \frac{1}{2}x(t)$ **C.** $\frac{1}{2}, -\frac{1}{2}x(t)$ **D.** $-\frac{1}{2}, -\frac{1}{2}x(t)$

Q1.01.1 MCQ 2001 1M

If a signal $f(t)$ has energy E, the energy of the signal $f(2t)$ is equal to

A. E **B.** $E/2$ **C.** $2E$ **D.** $4E$

Q1.01.2 MCQ 2001 1M

Let $\delta(t)$ denote the delta function. The value of the integral $\int_{-\infty}^{\infty} \delta(t)\cos\left(\frac{3t}{2}\right)dt$ is

A. 1 **B.** -1 **C.** 0 **D.** $\frac{\pi}{2}$

Q1.95.1 MCQ 1995 1M

The RMS value of a rectangular wave of period T, having a value of $+V$ for a duration, $T_1 (< T)$ and $-V$ for the duration, $T - T_1 = T_2$, equals

A. V **B.** $\frac{T_1 - T_2}{T}V$
C. $\frac{V}{\sqrt{2}}$ **D.** $\frac{T_1}{T_2}V$

Q1.92.1 MSQ 1992 2M

Which of the following signals is/are periodic?

- ☐ **A.** $S(t) = \cos 2t + \cos 3t + \cos 5t$
- ☐ **B.** $S(t) = \exp(j8\pi t)$
- ☐ **C.** $S(t) = \exp(-7t)\sin 10\pi t$
- ☐ **D.** $S(t) = \cos 2t \cos 4t$

Answer Key

Q1.26.1 C	**Q1.26.2** A,C,D	**Q1.24.1** A,D|A,C,D
Q1.19.1 12	**Q1.15.1** 2	**Q1.14.1** D
Q1.13.1 A	**Q1.09.1** B	**Q1.06.1** D
Q1.05.1 A	**Q1.05.2** A	**Q1.01.1** B
Q1.01.2 A	**Q1.95.1** A	**Q1.92.1** A,B,D

UNIT

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Unit Snapshot*

Stream: EC	**Questions:** 36	**MCQ:** 29	**MSQ:** 1	**NAT:** 6
1M: 20	**2M:** 16	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–3M(2017)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 2M

Consider the discrete time system (S) with input $x[n]$ and output $y[n]$ as shown in the Figure. The two sub-systems represented by their impulse responses $h_1[n]$ and $h_2[n]$ are linear and time invariant.
Which of the following statements is necessarily TRUE?

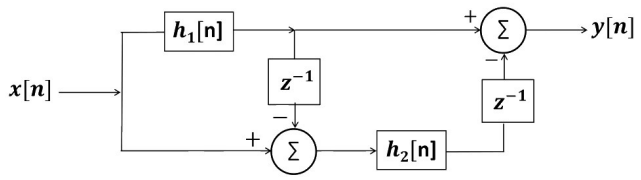


Fig. Q2.26.1

A. S is causal.

B. S is linear and time invariant.

C. S is linear and time varying.

D. S is non-linear.

Q2.25.1 MCQ 2025 1M

Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $b > 0$. Assuming $h_2[n] \neq \delta[n]$, the overall system (denoted by the dashed box) is _____.

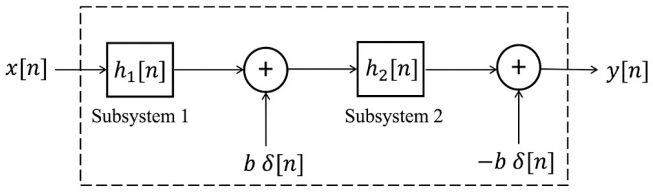


Fig. Q2.25.1

A. Linear and time invariant

B. Linear and time variant

C. Non linear and time invariant

D. Non linear and time variant

Q2.23.1 MCQ 2023 1M

Consider a system with input $x(t)$ and output $y(t) = x(e^t)$. The system is

A. Causal and time invariant

B. Non-causal and time varying

C. Causal and time varying

D. Non-causal and time invariant

Q2.22.1 MSQ 2022 2M

The outputs of four systems (S_1, S_2, S_3, S_4) corresponding to the input signal $\sin(t)$, for all time t, are shown in the figure. Based on the given information, which of the four systems is/are definitely NOT LTI (linear and time-invariant)?

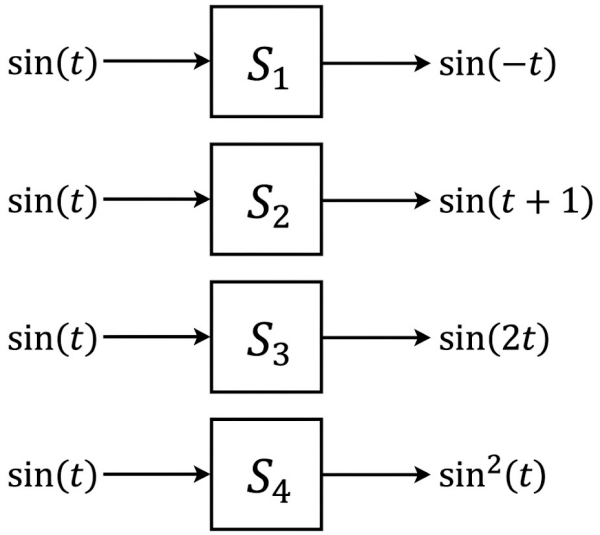


Fig. Q2.22.1

☐ **A.** S_1

☐ **B.** S_2

☐ **C.** S_3

☐ **D.** S_4

Q2.21.1 NAT 2021 2M

For a unit step input $u[n]$, a discrete-time LTI system produces an output signal $(2\delta[n+1] + \delta[n] + \delta[n-1])$. Let $y[n]$ be the output of the system for an input $\left(\left(\frac{1}{2}\right)^n u[n]\right)$. The value of $y[0]$ is _____.

Q2.20.1 MCQ 2020 1M

The output $y[n]$ of a discrete-time system for an input $x[n]$ is

$$y[n] = \max_{-\infty \le k \le n} |x[k]|.$$

The unit impulse response of the system is

A. unit step signal $u[n]$.

B. unit impulse signal $\delta[n]$.

C. 1 for all n.

D. 0 for all n.

Q2.18.1 MCQ 2018 1M

Let the input be u and the output be y of a system, and the other parameters are real constants. Identify which among the following systems is not a linear system:

A. $\frac{d^3y}{dt^3} + a_1\frac{d^2y}{dt^2} + a_2\frac{dy}{dt} + a_3y = b_3u + b_2\frac{du}{dt} + b_1\frac{d^2u}{dt^2}$ (with initial rest conditions)

B. $y(t) = \int_0^t a^{\alpha(t-\tau)}\beta u(\tau)d\tau$

C. $y = au + b, b \neq 0$

D. $y = au$

Q2.17.1 MCQ 2017 1M

The input $x(t)$ and the output $y(t)$ of a continuous time system are related as

$$y(t) = \int_{t-T}^{t} x(u)du$$

The system is

A. Linear and time variant

B. linear and time invariant

C. Non linear and time variant

D. Nonlinear and time invariant

Q2.17.2 MCQ 2017 1M

Consider a single input single output discrete-time system with $x[n]$ as input and $y[n]$ as output, where the two are related as

$$y[n] = \begin{cases} n|x[n]|, & \text{for } 0 \le n \le 10 \\ x[n] - x[n-1], & \text{otherwise,} \end{cases}$$

Which one of the following statements is true about the system?

A. It is causal and stable

B. It is causal but not stable

C. It is not causal but stable

D. It is neither causal nor stable

Q2.17.3 NAT 2017 2M

Consider the parallel combination of two LTI systems shown in the figure.

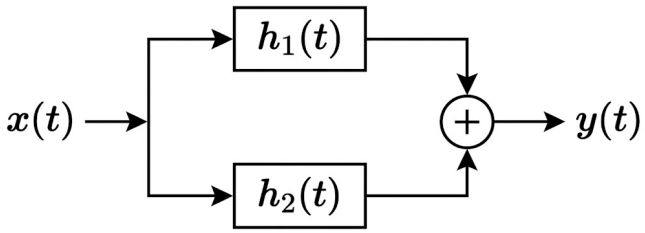


Fig. Q2.17.3

The impulse responses of the systems are

$$h_1(t) = 2\delta(t+2) - 3\delta(t+1)$$

$$h_2(t) = \delta(t-2)$$

If the input $x(t)$ is a unit step signal, then the energy of $y(t)$ is _____.

Q2.17.4 NAT 2017 2M

Two discrete-time signals $x[n]$ and $h[n]$ are both non-zero for $n = 0, 1, 2$ and are zero otherwise. It is given that $x[0] = 1, x[1] = 2, x[2] = 1, h[0] = 1$. Let $y[n]$ be the linear convolution of $x[n]$ and $h[n]$. Given that $y[1] = 3$ and $y[2] = 4$, the value of the expression $10y[3] + y[4]$ is _____.

Q2.15.1 MCQ 2015 1M

The impulse response of an LTI system can be obtained by

A. Differentiating the unit ramp response

B. Differentiating the unit step response

C. Integrating the unit ramp response

D. Integrating the unit step response

Q2.15.2 MCQ 2015 1M

The result of the convolution $x(-t) * \delta(-t - t_o)$ is

A. $x(t + t_0)$

B. $x(t - t_0)$

C. $x(-t + t_0)$

D. $x(-t - t_0)$

Q2.15.3 NAT 2015 2M

Two sequences $x_1[n]$ and $x_2[n]$ have the same energy. Suppose $x_1[n] = \alpha 0.5^n u[n]$, where α is a positive real number and $u[n]$ is the unit step sequence. Assume

$$x_2[n] = \begin{cases} \sqrt{1.5} & \text{for } n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the value of α is _____.

Q2.14.1 NAT 2014 2M

Consider a discrete-time signal

$$x[n] = \begin{cases} n & \text{for } 0 \le n \le 10 \\ 0 & \text{otherwise} \end{cases}$$

If $y[n]$ is the convolution of $x[n]$ with itself, the value of $y[4]$ is _____.

Q2.13.1 MCQ 2013 1M

Two systems with impulse responses $h_1(t)$ and $h_2(t)$ are connected in cascade. Then the overall impulse response of the cascaded system is given by

A. Product of $h_1(t)$ and $h_2(t)$

B. Sum of $h_1(t)$ and $h_2(t)$

C. Convolution of $h_1(t)$ and $h_2(t)$

D. Subtraction of $h_2(t)$ from $h_1(t)$

Q2.13.2 MCQ 2013 1M

The impulse response of a continuous time system is given by $h(t) = \delta(t-1)+\delta(t-3)$. The value of the step response at $t = 2$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q2.12.1 MCQ 2012 2M

The input $x(t)$ and output $y(t)$ of a system are related as $y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)d\tau$. The system is

A. Time-invariant and stable

B. Stable and not time-invariant

C. Time-invariant and not stable

D. Not time-invariant and not stable

Q2.12.2 MCQ 2012 1M

Let $y[n]$ denote the convolution of $h[n]$ and $g[n]$, where $h[n] = (1/2)^n u[n]$ and $g[n]$ is a causal sequence. If $y[0] = 1$ and $y[1] = 1/2$, then $g[1]$ equals

A. 0 **B.** 1/2 **C.** 1 **D.** 3/2

Q2.11.1 MCQ 2011 1M

A system is defined by its impulse response $h(n) = 2^n u(n-2)$. The system is

A. Stable and causal

B. Causal but not stable

C. Stable but not causal

D. Unstable and non-causal

Q2.10.1 MCQ 2010 1M

Two discrete time systems with impulse responses $h_1[n] = \delta[n-1]$ and $h_2[n] = \delta[n-2]$ are connected in cascade. The overall impulse response of the cascaded system is

A. $\delta[n-1] + \delta[n-2]$ **B.** $\delta[n-4]$
C. $\delta[n-3]$ **D.** $\delta[n-1]\delta[n-2]$

Q2.09.1 MCQ 2009 2M

Consider a system whose input x and output y are related by the equation

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(2\tau)d\tau,$$

where $h(t)$ is shown in the graph.

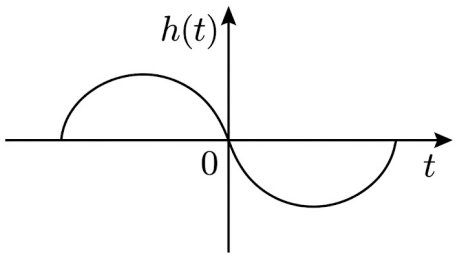


Fig. Q2.09.1

Which of the following four properties are possessed by the system ?

BIBO: Bounded input gives a bounded output.
Causal: The system is causal.
LP: The system is low pass.
LTI: The system is linear and time-invariant.

A. Causal, LP
B. BIBO, LTI
C. BIBO, Causal, LTI
D. LP, LTI

Q2.08.1 MCQ 2008 1M

The input and output of a continuous time system are respectively denoted by $x(t)$ and $y(t)$. Which of the following descriptions corresponds to a causal system?

A. $y(t) = x(t-2) + x(t+4)$

B. $y(t) = (t-4)x(t+1)$

C. $y(t) = (t+4)x(t-1)$

D. $y(t) = (t+5)x(t+5)$

Q2.08.2 MCQ 2008 1M

The impulse response $h(t)$ of a linear time invariant continuous time system is described by $h(t) = \exp(\alpha t)u(t) + \exp(\beta t)u(-t)$, where $u(t)$ denotes the unit step function, and α and β are real constants. This system is stable if

A. α is positive and β is positive

B. α is negative and β is negative

C. α is positive and β is negative

D. α is negative and β is positive

Q2.08.3 MCQ 2008 2M

Let $x(t)$ be the input and $y(t)$ be the output of a con tinuous time system. Match the system properties P1, P2 and P3 with system relations R1, R2, R3, R4.

Properties	**Relations**
P1: Linear but NOT time-invariant	R1: $y(t) = t^2x(t)$
P2: Time-invariant but NOT linear	R2 : $y(t) = t\|x(t)\|$
P3: Linear and time-invariant	R3 : $y(t) = \|x(t)\|$
	R4 : $y(t) = x(t-5)$

A. (P1, R1), (P2, R3), (P3, R4)

B. (P1, R2), (P2, R3), (P3, R4)

C. (P1, R3), (P2, R1), (P3, R2)

D. (P1, R1), (P2, R2), (P3, R3)

Q2.06.1 MCQ 2006 2M

A system with input $x[n]$ and output $y[n]$ is given as $y[n] = \left(\sin \frac{5}{6}\pi n\right) x[n]$. The system is:

A. Linear, stable and invertible

B. Non-linear, stable and non-invertible

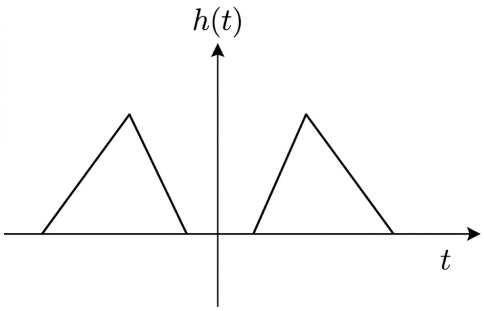
C. Linear, stable and non-invertible

D. Linear, unstable and invertible

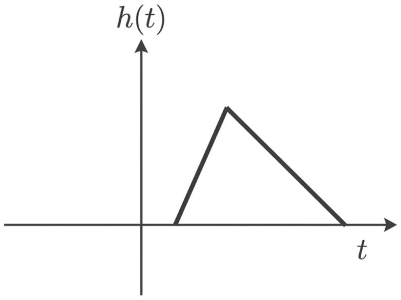
Q2.05.1 MCQ 2005 1M

Which of the following can be impulse response of a causal system ?

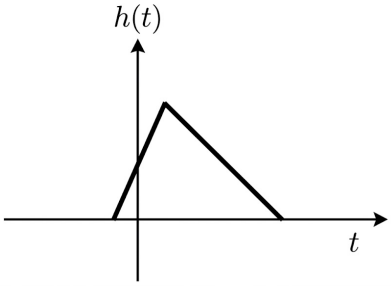
A.



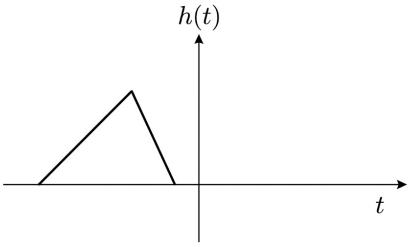
B.



C.



D.



Q2.04.1 MCQ 2004 1M

The impulse response $h[n]$ of a linear time-invariant system is given by $h[n] = u[n+3] + u[n-2] - 2u[n-7]$ where $u[n]$ is the unit step sequence. The above system is

A. Stable but not causal

B. Stable and causal

C. Causal but unstable

D. Unstable and not causal

Q2.03.1 MCQ 2003 2M

Let P be linearity, Q be time-invariance, R be causality and S be stability. A discrete time system has the input-output relationship,

$$y(n) = \begin{cases} x(n) & n \geq 1 \\ 0, & n = 0 \\ x(n+1) & n \leq -1 \end{cases}$$

where $x(n)$ is the input and $y(n)$ is the output. The above system has properties

A. P, S but not Q, R

B. P, Q, S but not R

C. P, Q, R, S

D. Q, R, S but not P

Q2.02.1 MCQ 2002 1M

Convolution of $x(t+5)$ with impulse function $\delta(t-7)$ is equal to

A. $x(t-12)$

B. $x(t+12)$

C. $x(t-2)$

D. $x(t+2)$

Q2.01.1 MCQ 2001 2M

The impulse response functions of four linear systems S1, S2, S3, S4 are given respectively by

$h_1(t) = 1$

$h_2(t) = u(t)$

$h_3(t) = \frac{u(t)}{t+1}$

$h_4(t) = e^{-3t}u(t)$

where $u(t)$ is the step function. Which of these systems is time invariant, causal and stable?

A. S1 **B.** S2
C. S3 **D.** S4

Q2.00.1 MCQ 2000 1M

A system with an input $x(t)$ and output $y(t)$ is described by the relation: $y(t) = tx(t)$. This system is

A. Linear and time-invariant

B. Linear and time varying

C. Non-linear and time-invariant

D. Non-linear and time-varying

Q2.95.1 MCQ 1995 1M

Let $h(t)$ be the impulse response of a linear time invariant system. Then the response of the system for any input $u(t)$ is

A. $\int_0^t h(\tau)u(t-\tau)d\tau$ **B.** $\frac{d}{dt}\int_0^t h(\tau)u(t-\tau)d\tau$

C. $|\int_0^t h(\tau)u(t-\tau)d\tau|$ **D.** $\int_0^t h^2(\tau)u(t-\tau)d\tau$

Q2.94.1 NAT 1994 2M

Match each of the items A, B and C with an appropriate item from 1, 2, 3, 4 and 5.

(A) $a_1\frac{d^2y}{dx^2} + a_2y\frac{dy}{dx} + a_3y = a_4$

(B) $a_1\frac{d^3y}{dx^3} + a_2y = a_3$

(C) $a_1\frac{d^2y}{dx^2} + a_2x\frac{dy}{dx} + a_3x^2y = 0$

(1) Non-linear differential equation

(2) Linear differential equation with constant coefficients

(3) Linear homogeneous differential equation

(4) Non-linear homogeneous differential equation

(5) Non-linear first order differential equation

Q2.91.1 MCQ 1991 2M

An excitation is applied to a system at $t = T$ and its response is zero for $-\infty < t < T$. Such a system is a

A. Non-causal system

B. Stable system

C. Causal system

D. Unstable system

Q2.90.1 MCQ 1990 2M

The impulse response and the excitation function of a linear time invariant casual system are shown in Fig. a and b respectively. The output of the system at $t = 2$ sec. is equal to

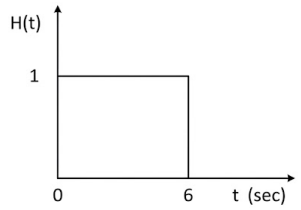


Fig. a

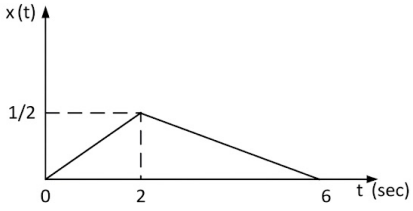


Fig. b

A. 0

B. 1/2

C. 3/2

D. 1

Answer Key

Q2.26.1 B	Q2.25.1 D	Q2.23.1 B
Q2.22.1 C,D	Q2.21.1 0	Q2.20.1 A
Q2.18.1 C	Q2.17.1 B	Q2.17.2 A
Q2.17.3 7	Q2.17.4 31	Q2.15.1 B
Q2.15.2 D	Q2.15.3 1.5	Q2.14.1 10
Q2.13.1 C	Q2.13.2 B	Q2.12.1 D
Q2.12.2 A	Q2.11.1 B	Q2.10.1 C
Q2.09.1 B	Q2.08.1 C	Q2.08.2 D
Q2.08.3 A	Q2.06.1 C	Q2.05.1 B
Q2.04.1 A	Q2.03.1 A	Q2.02.1 C
Q2.01.1 D	Q2.00.1 B	Q2.95.1 A
Q2.94.1 A-1, B-2, C-3	Q2.91.1 C	Q2.90.1 B

PrepFusion

UNIT

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 24	**MCQ:** 20	**MSQ:** 1	**NAT:** 3
1M: 15	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2015)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

The continuous time signal $x(t)$ is real, periodic with period T and satisfies the Dirichlet conditions.
The Fourier series representation of

$$x(t) = \sum_{-\infty}^{\infty} a_n e^{j\left(\frac{2\pi nt}{T}\right)}$$

and $x(t)$ satisfies the following: $x\left(t - \frac{T}{2}\right) = -x(t)$.
For any integer m, which of the following options is correct?

A. $a_{2m} = 0$

B. $a_{2m} = 1$

C. $a_{2m} = a_{2m+1}$

D. $a_{2m} = -1$

Q3.25.1 MSQ 2025 2M

Let $f(t)$ be a periodic signal with fundamental period $T_0 > 0$. Consider the signal $y(t) = f(\alpha t)$, where $\alpha > 1$.
The Fourier series expansions of $f(t)$ and $y(t)$ are given by

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\frac{2\pi}{T_0}kt} \quad \text{and} \quad y(t) = \sum_{k=-\infty}^{\infty} d_k e^{j\frac{2\pi}{T_0}\alpha kt}.$$

Which of the following statements is/are TRUE?

☐ **A.** $c_k = d_k$ for all k

☐ **B.** $y(t)$ is periodic with a fundamental period αT_0

☐ **C.** $c_k = d_k/\alpha$ for all k

☐ **D.** $y(t)$ is periodic with a fundamental period T_0/α

Q3.21.1 NAT 2021 2M

The exponential Fourier series representation of a continuous-time periodic signal $x(t)$ is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_o t}$$

where ω_o is the fundamental angular frequency of $x(t)$ and the coefficients of the series are a_k. The following information is given about $x(t)$ and a_k.

I. $x(t)$ is real and even, having a fundamental period of 6.

II. The average value of $x(t)$ is 2.

III. $a_k = \begin{cases} k, & 1 \le k \le 3 \\ 0, & k > 3 \end{cases}$

The average power of the signal $x(t)$ (rounded off to one decimal place) is _____.

Q3.18.1 MCQ 2018 1M

Let $x(t)$ be a periodic function with period $T = 10$. The Fourier series coefficients for this series are denoted by a_k, that is

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$

The same function $x(t)$ can also be considered as periodic function with period $T' = 40$. Let b_k be the Fourier series coefficients when period is taken as T'.
If $\sum_{k=-\infty}^{\infty} |a_k| = 16$, then $\sum_{k=-\infty}^{\infty} |b_k|$ is equal to

A. 256 **B.** 64 **C.** 16 **D.** 4

Q3.17.1 MCQ 2017 1M

A periodic signal $x(t)$ has a trigonometric Fourier series expansion

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

If $x(t) = -x(-t) = -x(t - \pi/\omega_0)$, we can conclude that

A. a_n are zero for all n and b_n are zero for n even

B. a_n are zero for all n and b_n are zero for n odd

C. a_n are zero for n even and b_n are zero for n odd

D. a_n are zero for n odd and b_n are zero for n even

Q3.17.2 MCQ 2017 2M

Let $x(t)$ be a continuous time periodic signal with fundamental period $T = 1$ seconds. Let $\{a_k\}$ be the complex Fourier series coefficients of $x(t)$, where k is integer valued. Consider the following statements about $x(3t)$:
I. The complex Fourier series coefficients of $x(3t)$ are $\{a_k\}$ where k is integer valued.
II. The complex Fourier series coefficients of $x(3t)$ are $\{3a_k\}$ where k is integer valued.
III. The fundamental angular frequency of $x(3t)$ is 6π rad/s.
For the three statements above, which one of the following is correct?

A. only II and III are true **B.** only I and III are true
C. only III is true **D.** only I is true

Q3.16.1 MCQ 2016 2M

A signal $2\cos\left(\frac{2\pi}{3}t\right) - \cos(\pi t)$ is the input to an LTI system with the transfer function

$$H(s) = e^{s} + e^{-s}.$$

If C_k denotes the kth coefficient in the exponential Fourier series of the output signal, then C_3 is equal to

A. 0 **B.** 1 **C.** 2 **D.** 3

Q3.15.1 MCQ 2015 1M

The magnitude and phase of the complex Fourier series coefficients a_k of a periodic signal $x(t)$ are shown in the figure. Choose the correct statement from the four choices given. Notation: C is the set of complex numbers, R is the set of purely real numbers, and P is the set of purely imaginary numbers.

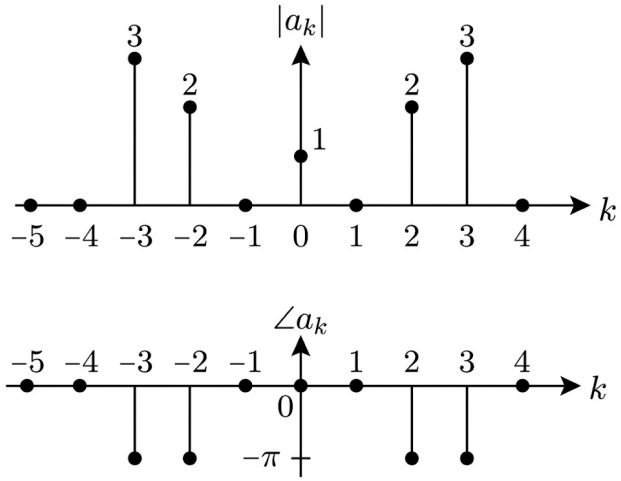


Fig. Q3.15.1

A. $x(t) \in R$

B. $x(t) \in P$

C. $x(t) \in (C - R)$

D. the information given is not sufficient to draw any conclusion about $x(t)$

Q3.14.1 NAT 2014 1M

Consider the periodic square wave in the figure shown.

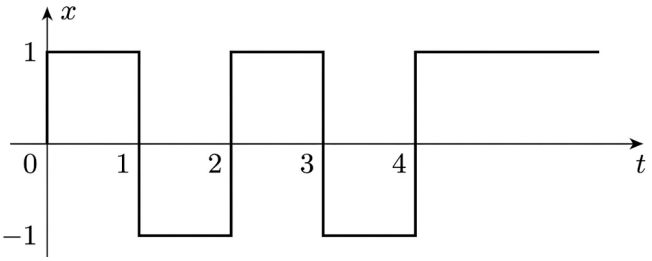


Fig. Q3.14.1

The ratio of the power in the 7th harmonic to the power in the 5th harmonic for this waveform is closest in value to _____.

Q3.11.1 MCQ 2011 1M

The trigonometric Fourier series of an even function does not have the

A. dc term

B. cosine terms

C. sine terms

D. odd harmonic terms

Q3.10.1 MCQ 2010 1M

The trigonometric Fourier series for the waveform $f(t)$ shown below contains

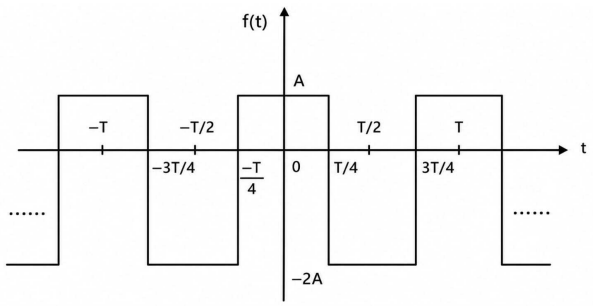


Fig. Q3.10.1

A. only cosine terms and zero values for the dc components

B. only cosine terms and a positive value for the dc components

C. only cosine terms and a negative value for the dc components

D. only sine terms and a negative value for the dc components

Q3.09.1 MCQ 2009 1M

The Fourier series of a real periodic function has only
(P) cosine terms if it is even
(Q) sine terms if it is even
(R) cosine terms if it is odd
(S) sine terms if it is odd
Which of the above statements are correct ?

A. P and S

B. P and R

C. Q and S

D. Q and R

Q3.05.1 MCQ 2005 1M

Choose the function $f(t); -\infty < t < \infty$ for which a Fourier series cannot be defined.

A. $3\sin(25t)$

B. $4\cos(20t + 3) + 2\sin(10t)$

C. $\exp(-|t|)\sin(25t)$

D. 1

Q3.04.1 MCQ 2004 2M

A rectangular pulse train $s(t)$ as shown in the figure is convolved with the signal $\cos^2(4\pi \times 10^3 t)$. The convolved signal will be a

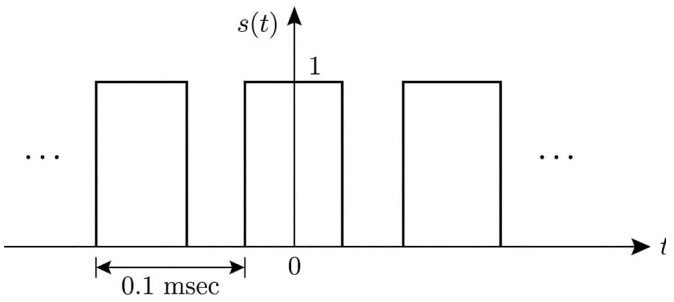


Fig. Q3.04.1

A. DC

B. 12 kHz sinusoid

C. 8 kHz sinusoid

D. 14 kHz sinusoid

Q3.03.1 MCQ 2003 1M

The Fourier series expansion of a real periodic signal with fundamental frequency f_0 is given by

$$g_p(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t}$$

It is given that $c_3 = 3 + j5$. Then c_{-3} is

A. $5 + j3$

B. $-3 - j5$

C. $-5 + j3$

D. $3 - j5$

Q3.02.1 MCQ 2002 1M

Which of the following cannot be the Fourier series expansion of a periodic signal?

A. $x(t) = 2\cos t + 3\cos 3t$

B. $x(t) = 2\cos \pi t + 7\cos t$

C. $x(t) = \cos t + 0.5$

D. $x(t) = 2\cos 1.5\pi t + \sin 3.5\pi t$

Q3.00.1 MCQ 2000 1M

One period (0,T) each of two periodic waveforms, W_1 and W_2 are shown in Fig.Q3.00.1. The magnitudes of the nth Fourier series coefficients of W_1 and W_2, for $n \geq 1$, n odd, are respectively proportional to

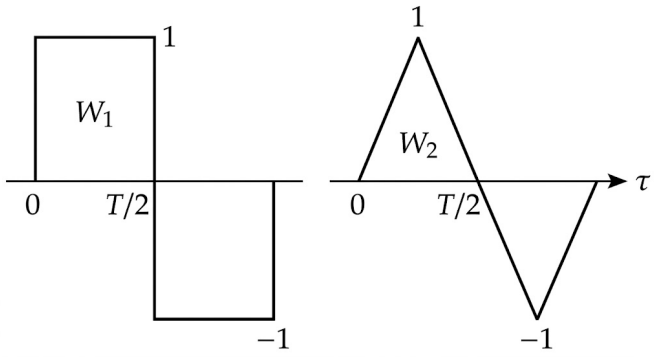


Fig. Q3.00.1

A. $|n^{-3}|$ and $|n^{-2}|$

B. $|n^{-2}|$ and $|n^{-3}|$

C. $|n^{-1}|$ and $|n^{-2}|$

D. $|n^{-4}|$ and $|n^{-2}|$

Q3.99.1 MCQ 1999 2M

The Fourier series representation of an impulses train denoted by

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

is given by

A. $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} \exp(-\frac{j2\pi nt}{T_0})$

B. $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} \exp(-\frac{j\pi nt}{T_0})$

C. $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} \exp(\frac{j\pi nt}{T_0})$

D. $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} \exp(\frac{j2\pi nt}{T_0})$

Q3.98.1 MCQ 1998 1M

A periodic signal $x(t)$ of period T_o is given by

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < \frac{T_o}{2} \end{cases}$$

The d.c. component of $x(t)$ is

A. $\frac{T_1}{T_0}$ **B.** $\frac{T_1}{2T_0}$ **C.** $\frac{2T_1}{T_0}$ **D.** $\frac{T_0}{T_1}$

Q3.98.2 MCQ 1998 1M

The trigonometric Fourier series of a periodic time function can have only

A. cosine terms **B.** sine terms
C. cosine and sine terms **D.** d.c. and cosine terms

Q3.96.1 MCQ 1996 1M

The trigonometric Fourier series of an even function of time does not have the

A. dc term **B.** cosine terms
C. sine terms **D.** odd harmonic terms

Q3.94.1 MCQ 1994 1M

The Fourier Series of a odd periodic function, contains only

A. odd harmonics **B.** even harmonics
C. cosine terms **D.** sine terms

Q3.93.1 NAT 1993 2M

Fourier series of the periodic function (period 2π) defined by

$$f(x) = \begin{cases} 0 & -\pi \leqslant x < 0 \\ x & 0 < x < \pi \end{cases} \text{ is}$$

$$\frac{\pi}{4} + \sum_{1}^{\infty}\left[\frac{1}{\pi n^2}(\cos n\pi - 1)\cos nx - \frac{1}{n}\cos n\pi \sin nx\right]$$

By putting $x = \pi$ in the above, one can deduce that the sum of the series

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots, \text{ is}$$

Q3.87.1 MCQ 1987 2M

A half-wave rectified sinusoidal waveform has a peak voltage of 10 V. Its average value and the peak value of the fundamental component are respectively given by

A. $\frac{20}{\pi}$ V, $\frac{10}{\sqrt{2}}$ V **B.** $\frac{10}{\pi}$ V, $\frac{10}{\sqrt{2}}$ V
C. $\frac{10}{\pi}$ V, 5 V **D.** $\frac{20}{\pi}$ V, 5 V

Answer Key

Q3.26.1 A	Q3.25.1 A,D	Q3.21.1 32
Q3.18.1 C	Q3.17.1 A	Q3.17.2 B
Q3.16.1 B	Q3.15.1 A	Q3.14.1 0.51
Q3.11.1 C	Q3.10.1 C	Q3.09.1 A
Q3.05.1 C	Q3.04.1 A	Q3.03.1 D
Q3.02.1 B	Q3.00.1 C	Q3.99.1 D
Q3.98.1 C	Q3.98.2 C	Q3.96.1 C
Q3.94.1 D	Q3.93.1 $\frac{\pi^2}{8}$	Q3.87.1 C

UNIT

IV

Continuous Time Fourier Transform

Topics

1. Introduction to Fourier Transform
2. Properties of Fourier Transform
3. Fourier Transform of Standard Signals
4. Duality Property
5. FT of Rectangular, Triangular and Sampling Signals
6. Fourier Transform of Periodic Signals
7. Differentiation, Integration, Multiplication and Convolution Property
8. Parseval's Theorem
9. LTI Systems and Analog Filters
10. Nyquist Rate - Baseband and Bandpass Signals
11. Sampling Theorem and Applications

Unit Snapshot*

Stream: EC	**Questions:** 69	**MCQ:** 56	**MSQ:** 0	**NAT:** 13
1M: 33	**2M:** 36	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2022)–6M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M

Two analog signals $x_1(t)$ and $x_2(t)$ (t in second), are sampled at a rate $F_s = 40$ Hz, where $x_1(t) = \cos(20\pi t)$, $t \geq 0$ and $x_2(t) = \cos(100\pi t), t \geq 0$.
The first ten samples (starting from $t = 0$) are considered for the analysis. Which of the following statements is TRUE?

A. All of the first three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.

B. All of the last three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.

C. All of the samples of $x_2(t)$ are greater than the corresponding samples of $x_1(t)$.

D. All of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

Q4.26.2 MCQ 2026 2M

Consider a real baseband signal $x(t) = e^{-2t}$, for t (in seconds) ≥ 0.
If 99% of energy of $x(t)$ lies within B Hz, then which of the following options is TRUE for the value of B?

A. $B > 1$ kHz

B. $\frac{63}{\pi}$ Hz $< B < \frac{64}{\pi}$ Hz

C. $\frac{126}{\pi}$ Hz $< B < \frac{128}{\pi}$ Hz

D. $B < 1$ Hz

Q4.25.1 MCQ 2025 1M

Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t)\exp(-j\omega t)dt$ exists. Which one of the following statements is always TRUE?

A. $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|dt$

B. $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)|dt$

C. $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t)dt$

D. $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t)dt$

Q4.25.2 MCQ 2025 2M

Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec. The signal $f(t)$ is uniformly sampled to obtain $y(t) = f(t)p(t)$. Here

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s)$$

with $\delta(t)$ being the Dirac impulse, $T_s > 0$, and $\tau > 0$. The sampled signal $y(t)$ is passed through an ideal low pass filter $h(t) = \omega_c T_s \frac{\sin(\omega_c t)}{\pi \omega_c t}$ with cutoff frequency ω_c and passband gain T_s. The output of the filter is given by .

A. $f(t)$ if $T_s < \pi/\omega_c$

B. $f(t - \tau)$ if $T_s < \pi/\omega_c$

C. $f(t - \tau)$ if $T_s < 2\pi/\omega_c$

D. $T_s f(t)$ if $T_s < 2\pi/\omega_c$

Q4.24.1 MCQ 2024 2M

A continuous time signal $x(t) = 2\cos(8\pi t + \pi/3)$ is sampled at a rate of 15 Hz. The sampled signal $x_s(t)$ when passed through an LTI system with impulse response

$$h(t) = \left(\frac{\sin 2\pi t}{\pi t}\right)\cos(38\pi t - \pi/2)$$

produces an output $x_0(t)$. The expression for $x_0(t)$ is _____

A. $15\sin(38\pi t + \pi/3)$

B. $15\sin(38\pi t - \pi/3)$

C. $15\cos(38\pi t - \pi/6)$

D. $15\cos(38\pi t + \pi/6)$

Q4.24.2 MCQ 2024 2M

Consider two continuous time signals $x(t)$ and $y(t)$ as shown below

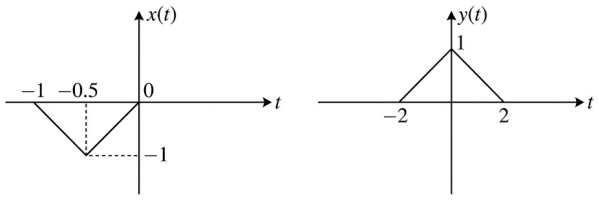


Fig. Q4.24.2

If $X(f)$ denotes the Fourier transform of $x(t)$, then the Fourier transform of $y(t)$ is ___

A. $-4X(4f)e^{-j\pi f}$

B. $-4X(4f)e^{-j4\pi f}$

C. $-\frac{1}{4}X(f/4)e^{-j\pi f}$

D. $-\frac{1}{4}X(f/4)e^{-j4\pi f}$

Q4.23.1 MCQ 2023 1M

The Fourier transform $X(\omega)$ of $x(t) = e^{-t^2}$ is

A. $\sqrt{\pi}e^{\frac{\omega^2}{2}}$ **B.** $\frac{e^{-\frac{\omega^2}{4}}}{2\sqrt{\pi}}$

C. $\sqrt{\pi}e^{-\frac{\omega^2}{4}}$ **D.** $\sqrt{\pi}e^{-\frac{\omega^2}{2}}$

Q4.23.2 NAT 2023 2M

Let $x_1(t) = u(t + 1.5) - u(t - 1.5)$ and $x_2(t)$ is shown in the figure below. For $y(t) = x_1(t) * x_2(t)$, the $\int_{-\infty}^{\infty} y(t)dt$ is ___. (rounded off to the nearest integer).

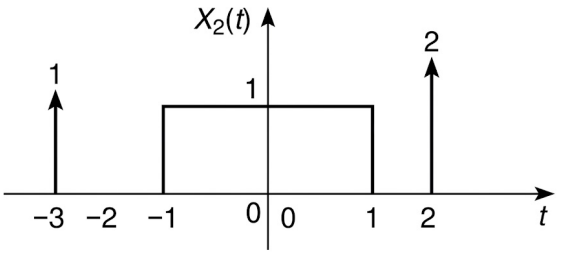


Fig. Q4.23.2

Q4.23.3 MCQ 2023 1M

Let $m(t)$ be a strictly band-limited signal with bandwidth B and energy E. Assuming $\omega_0 = 10B$, the energy in the signal $m(t)\cos\omega_0 t$ is

A. $\frac{E}{4}$ **B.** $\frac{E}{2}$

C. E **D.** $2E$

Q4.23.4 MCQ 2023 2M

Let $x(t) = 10\cos(10.5Wt)$ be passed through an LTI system having impulse response $h(t) = \pi\left(\frac{\sin Wt}{\pi t}\right)^2 \cos 10Wt$.The output of the system is

A. $\left(\frac{15W}{4}\right)\cos(10.5Wt)$ **B.** $\left(\frac{15W}{2}\right)\cos(10.5Wt)$

C. $\left(\frac{15W}{8}\right)\cos(10.5Wt)$ **D.** $(15W)\cos(10.5Wt)$

Q4.22.1 MCQ 2022 1M

The Fourier transform $X(j\omega)$ of the signal is _____.

$$x(t) = \frac{t}{(1+t^2)^2}$$

A. $\frac{\pi}{2j}\omega e^{-|\omega|}$ **B.** $\frac{\pi}{2}\omega e^{-|\omega|}$

C. $\frac{\pi}{2j}e^{-|\omega|}$ **D.** $\frac{\pi}{2}e^{-|\omega|}$

Q4.21.1 MCQ 2021 1M

Consider a real-valued base-band signal $x(t)$, band limited to 10 kHz. The Nyquist rate for the signal $y(t) = x(t)x\left(1 + \frac{t}{2}\right)$ is

A. 15 kHz **B.** 30 kHz

C. 60 kHz **D.** 20 kHz

Q4.20.1 NAT 2020 2M

$X(\omega)$ is the Fourier transform of $x(t)$ shown below. The value of $\int_{-\infty}^{\infty}|X(\omega)|^2 d\omega$ (rounded off to two decimal places) is _____.

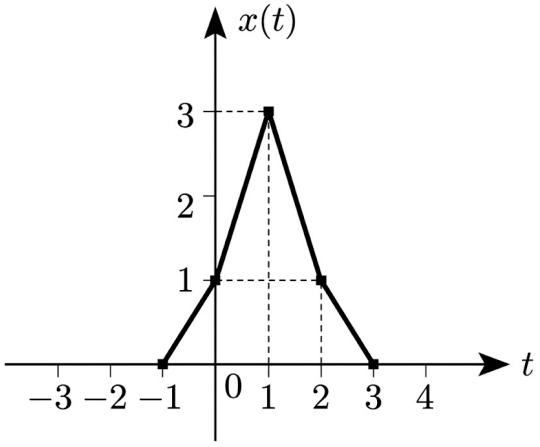


Fig. Q4.20.1

Q4.18.1 NAT 2018 2M

The input $4\text{sinc}(2t)$ is fed to a Hilbert transformer to obtain $y(t)$, as shown in the figure below:



Fig. Q4.18.1

Here $sinc(x) = \frac{\sin(\pi x)}{\pi x}$. The value (accurate to two

decimal places) of $\int_{-\infty}^{\infty} |y(t)|^2 dt$ is _____.

Q4.18.2 NAT 2018 2M

A band limited low-pass signal $x(t)$ of bandwidth 5 kHz is sampled at a sampling rate f_s. The signal $x(t)$ is reconstructed using the reconstruction filter $H(f)$ whose magnitude response is shown below:

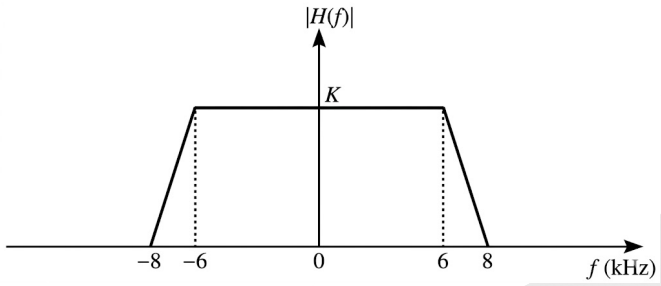


Fig. Q4.18.2

The minimum sampling rate f_s (in kHz) for perfect reconstruction of $x(t)$ is _____.

Q4.17.1 MCQ 2017 2M

The signal $x(t) = \sin(14000\pi t)$, where t is in seconds is sampled at a rate of 9000 samples per second. The sampled signal is the input to an ideal low pass filter with frequency response $H(f)$ as follows :

$$H(f) = \begin{cases} 1, & |f| \le 12 \text{ kHz} \\ 0, & |f| \ge 12 \text{ kHz} \end{cases}$$

What is the number of sinusoids in the output and their frequencies in kHz?

A. Number = 1, frequency = 7

B. Number = 3, frequencies = 2, 7, 11

C. Number = 2, frequencies = 2, 7

D. Number = 2, frequencies = 2, 11

Q4.17.2 NAT 2017 2M

Consider an LTI system with magnitude response

$$|H(f)| = \begin{cases} 1 - \frac{|f|}{20} & |f| \le 20 \\ 0 & |f| > 20 \end{cases}$$

And phase response $\text{Arg}\{H(f)\} = -2f$, If the input to the system is

$$x(t) = 8\cos\left(20\pi t + \frac{\pi}{4}\right) + 16\sin\left(40\pi t + \frac{\pi}{8}\right) + 24\cos\left(80\pi t + \frac{\pi}{16}\right)$$

Then the average power of the output signal $y(t)$ is

Q4.17.3 NAT 2017 2M

A continuous time signal $x(t) = 4\cos(200\pi t) + 8\cos(400\pi t)$, where t is in seconds, is the input to a linear time invariant (LTI) filter with the impulse response

$$h(t) = \begin{cases} \frac{2\sin(300\pi t)}{\pi t} & t \ne 0 \\ 600 & t = 0 \end{cases}$$

Let $y(t)$ be the output of this filter. The maximum value of $|y(t)|$ is _____.

Q4.16.1 NAT 2016 1M

A continuous-time sinusoid of frequency 33 Hz is multiplied with a periodic Dirac impulse train of frequency 46 Hz. The resulting signal is passed through an ideal analog low-pass filter with a cutoff frequency of 23 Hz. The fundamental frequency (in Hz) of the output is _____.

Q4.16.2 MCQ 2016 1M

If the signal $x(t) = \frac{\sin(t)}{\pi t} * \frac{\sin(t)}{\pi t}$ with $*$ denoting the convolution operation, then x(t) is equal to

A. $\frac{\sin(t)}{\pi t}$ **B.** $\frac{\sin(2t)}{2\pi t}$

C. $\frac{2\sin(t)}{\pi t}$ **D.** $\left(\frac{\sin(t)}{\pi t}\right)^2$

Q4.16.3 NAT 2016 1M

The energy of the signal $x(t) = \frac{\sin(4\pi t)}{4\pi t}$ is _____.

Q4.16.4 MCQ 2016 1M

A continuous -time function $x(t)$ is periodic with period T. The function is sampled uniformly with a sampling period T_s. In which one of the following cases is the sampled signal periodic ?

A. $T = \sqrt{2}T_s$ **B.** $T = 1.2T_s$ **C.** Always **D.** Never

Q4.16.5 MCQ 2016 2M

A network consisting of a finite number of linear resistor (R), inductor (L), and capacitor (C) elements, connected all in series or all in parallel, is excited with a source of the form

$$\sum_{k=1}^{3} a_k \cos(k\omega_0 t), where a_k \ne 0, \omega_0 \ne 0.$$

The source has nonzero impedance. Which one of the following is a possible form of the output measured across a resistor in the network?

A. $\sum_{k=1}^{3} b_k \cos(k\omega_0 t + \phi_k)$, where $b_k \neq a_k, \forall k$

B. $\sum_{k=1}^{4} b_k \cos(k\omega_0 t + \phi_k)$, where $b_k \neq 0, \forall k$

C. $\sum_{k=1}^{3} a_k \cos(k\omega_0 t + \phi_k)$

D. $\sum_{k=1}^{2} a_k \cos(k\omega_0 t + \phi_k)$

Q4.16.6 MCQ 2016 1M

Consider the signal $x(t) = \cos(6\pi t) + \sin(8\pi t)$, where t is in seconds. The Nyquist sampling rate (in samples/second) for the signal $y(t) = x(2t + 5)$ is

A. 8 **B.** 12
C. 16 **D.** 32

Q4.15.1 NAT 2015 2M

Consider a continuous-time signal defined as

$$x(t) = \left(\frac{\sin(\pi t/2)}{(\pi t/2)}\right) * \sum_{n=-\infty}^{\infty} \delta(t - 10n)$$

where '$*$' denotes the convolution operation and t is in seconds. The Nyquist sampling rate (in samples/sec) for $x(t)$ is _____.

Q4.15.2 MCQ 2015 1M

The signal $\cos\left(10\pi t + \frac{\pi}{4}\right)$ is ideally sampled at a sampling frequency of 15 Hz. The sampled signal is passed through a filter with impulse response $\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(40\pi t - \frac{\pi}{2}\right)$. The filter output is

A. $\frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)$

B. $\frac{15}{2}\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(10\pi t + \frac{\pi}{4}\right)$

C. $\frac{15}{2}\cos\left(10\pi t - \frac{\pi}{4}\right)$

D. $\frac{15}{2}\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(40\pi t - \frac{\pi}{4}\right)$

Q4.15.3 NAT 2015 2M

The value of the integral $\int_{-\infty}^{\infty} 12\cos(2\pi t)\frac{\sin(4\pi t)}{4\pi t}dt$ is _____.

Q4.14.1 MCQ 2014 1M

A real-valued signal $x(t)$ limited to the frequency band $|f| \leq \frac{W}{2}$ is passed through a linear time invariant system whose frequency response is

$$H(f) = \begin{cases} e^{-j4\pi f}, & |f| \leq \frac{W}{2} \\ 0, & |f| > \frac{W}{2} \end{cases}$$

The output of the system is

A. $x(t + 4)$ **B.** $x(t - 4)$
C. $x(t + 2)$ **D.** $x(t - 2)$

Q4.14.2 NAT 2014 2M

The value of the integral $\int_{-\infty}^{\infty} \text{sinc}^2(5t)dt$ is _____.

Q4.14.3 MCQ 2014 2M

For a function $g(t)$, it is given that $\int_{-\infty}^{+\infty} g(t)e^{-j\omega t}dt = \omega e^{-2\omega^2}$ for any real value ω. If $y(t) = \int_{-\infty}^{t} g(\tau)d\tau$, then $\int_{-\infty}^{+\infty} y(t)dt$ is

A. 0 **B.** $-j$
C. $-\frac{j}{2}$ **D.** $\frac{j}{2}$

Q4.14.4 MCQ 2014 1M

Let $x(t) = \cos(10\pi t) + \cos(30\pi t)$ be sampled at 20 Hz and reconstructed using an ideal low-pass filter with cut-off frequency of 20 Hz. The frequency/ frequencies present in the reconstructed signal is/are

A. 5 Hz and 15 Hz only

B. 10 Hz and 15 Hz only

C. 5 Hz, 10 Hz and 15 Hz only

D. 5 Hz only

Q4.14.5 NAT 2014 1M

Consider two real valued signals, $x(t)$ band-limited to $[-500\text{ Hz}, 500\text{ Hz}]$ and $y(t)$ band-limited to $[-1\text{ kHz}, 1\text{ kHz}]$. For $z(t) = x(t) \cdot y(t)$, the Nyquist sampling frequency (in kHz) is _____.

Q4.13.1 MCQ 2013 1M

A band-limited signal with a maximum frequency of 5 kHz is to be sampled. According to the sampling theorem, the sampling frequency which is not valid is

A. 5 kHz **B.** 12 kHz
C. 15 kHz **D.** 20 kHz

Q4.12.1 MCQ 2012 2M

The Fourier transform of a signal $h(t)$ is

$$H(j\omega) = (2\cos\omega)(\sin 2\omega)/\omega$$

. The value of $h(0)$ is

A. $1/4$ **B.** $1/2$ **C.** 1 **D.** 2

Q4.10.1 MCQ 2010 2M

The Nyquist sampling rate for the signal

$$s(t) = \frac{\sin(500\pi t)}{\pi t} \times \frac{\sin(700)\pi t}{\pi t}$$

is given by

A. 400 Hz **B.** 600 Hz
C. 1200 Hz **D.** 1400 Hz

Q4.08.1 MCQ 2008 2M

The signal $x(t)$ is described by

$$x(t) = \begin{cases} 1 & for\ -1 \le t \le +1 \\ 0 & otherwise \end{cases}$$

Two of the angular frequencies at which its Fourier transform becomes zero are

A. $\pi, 2\pi$ **B.** $0.5\pi, 1.5\pi$
C. $0, \pi$ **D.** $2\pi, 2.5\pi$

Q4.08.2 MCQ 2008 2M

The impulse response $h(t)$ of a linear time invariant continuous time system is given by $h(t) = \exp(-2t)u(t)$, where $u(t)$ denotes the unit step function

The output of this system, to the sinusoidal input $x(t) = 2\cos(2t)$ for all time t, is

A. 0

B. $2^{-0.25}\cos(2t - 0.125\pi)$

C. $2^{-0.5}\cos(2t - 0.125\pi)$

D. $2^{-0.5}\cos(2t - 0.25\pi)$

Q4.08.3 MCQ 2008 2M

The impulse response $h(t)$ of a linear time invariant continuous time system is given by $h(t) = \exp(-2t)u(t)$, where $u(t)$ denotes the unit step function

The frequency response $H(\omega)$ of this system in terms of angular frequency ω, is given by $H(\omega) =$

A. $\frac{1}{1+j2\omega}$ **B.** $\frac{\sin\omega}{\omega}$
C. $\frac{1}{2+j\omega}$ **D.** $\frac{j\omega}{2+j\omega}$

Q4.06.1 MCQ 2006 1M

Let $x(t) \leftrightarrow X(j\omega)$ be Fourier Transform pair. The Fourier Transform of the signal $x(5t-3)$ in terms of $X(j\omega)$ is given as

A. $\frac{1}{5}e^{-j\frac{3\omega}{5}}X\left(\frac{j\omega}{5}\right)$ **B.** $\frac{1}{5}e^{j\frac{3\omega}{5}}X\left(\frac{j\omega}{5}\right)$
C. $\frac{1}{5}e^{-j3\omega}X\left(\frac{j\omega}{5}\right)$ **D.** $\frac{1}{5}e^{j3\omega}X\left(\frac{j\omega}{5}\right)$

Q4.06.2 MCQ 2006 2M

The minimum sampling frequency (in samples /sec) required to reconstruct the following signal from its samples without distortion $x(t) = 5\left(\frac{\sin\ 2\pi 1000t}{\pi t}\right)^3 + 7\left(\frac{\sin\ 2\pi 1000t}{\pi t}\right)^2$ would be

A. 2×10^3 **B.** 4×10^3
C. 6×10^3 **D.** 8×10^3

Q4.06.3 MCQ 2006 2M

A signal $m(t)$ with bandwidth 500 Hz is first multiplied by a signal $g(t)$ where

$g(t) = \sum_{k=-\infty}^{\infty}(-1)^k\delta(t - 0.5 \times 10^{-4}k)$

The resulting signal is then passed through an ideal lowpass filter with bandwidth 1 kHz. The output of the lowpass filter would be

A. $\delta(t)$ **B.** $m(t)$
C. 0 **D.** $m(t)\delta(t)$

Q4.05.1 MCQ 2005 2M

For a signal $x(t)$ the Fourier transform is $X(f)$. Then the inverse Fourier transform of $X(3f+2)$ is given by

A. $\frac{1}{2}x(\frac{t}{2})e^{-j\frac{4\pi}{2}t}$ **B.** $\frac{1}{3}x(\frac{t}{3})e^{-j\frac{4\pi}{3}t}$
C. $3x(3t)e^{-4j\pi t}$ **D.** $x(3t+2)$

Q4.05.2 MCQ 2005 1M

The output $y(t)$ of a linear time invariant system is related to its input $x(t)$ by the following equations:

$$y(t) = 0.5x(t - t_d + T) + x(t - t_d) + 0.5x(t - t_d - T)$$

The filter transfer function $H(\omega)$ of such a system is given by

A. $(1 + \cos\omega T)e^{-j\omega t_d}$

B. $(1 + 0.5\cos\omega T)e^{-j\omega t_d}$

C. $(1 + \cos\omega T)e^{j\omega t_d}$

D. $(1 - 0.5\cos\omega T)e^{-j\omega t_d}$

Q4.04.1 MCQ 2004 2M

Let $x(t)$ and $y(t)$ with Fourier transforms $X(f)$ and $Y(f)$ respectively be related as shown in Fig.

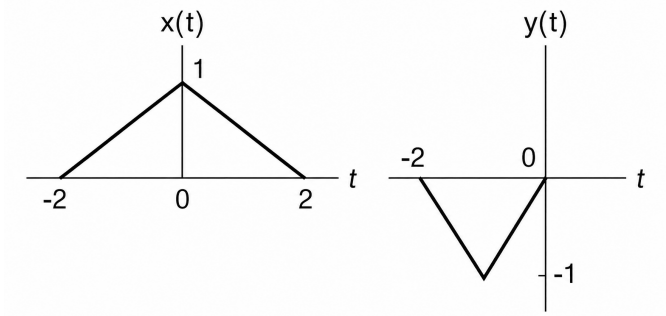


Fig. Q4.04.1

Then $Y(f)$ is

A. $-\frac{1}{2}X(f/2)e^{-j\pi f}$ **B.** $-\frac{1}{2}X(f/2)e^{j2\pi f}$
C. $-X(f/2)e^{j2\pi f}$ **D.** $-X(f/2)e^{-j2\pi f}$

Q4.04.2 MCQ 2004 1M

The Fourier transform of a conjugate symmetric function is always

A. imaginary **B.** conjugate anti-symmetric
C. real **D.** conjugate symmetric

Q4.04.3 MCQ 2004 2M

A 1 kHz sinusoidal signal is ideally sampled at 1500 samples/sec and the sampled signal is passed through an ideal low-pass filter with cut-off frequency 800 Hz. The output signal has the frequency.

A. zero Hz **B.** 0.75 kHz
C. 0.5 kHz **D.** 0.25 kHz

Q4.03.1 MCQ 2003 1M

Let $x(t)$ be the input to a linear, time-invariant system. The required output is $4x(t-2)$. The transfer function of the system should be

A. $4e^{j4\pi f}$ **B.** $2e^{-j8\pi f}$
C. $4e^{-j4\pi f}$ **D.** $2e^{j8\pi f}$

Q4.03.2 MCQ 2003 2M

Let $x(t) = 2\cos(800\pi t) + \cos(1400\pi t)$. $x(t)$ is sampled with the rectangular pulse train shown in the figure. The only spectral components (in kHz) present in the sampled signal in the frequency range 2.5 kHz to 3.5 kHz are

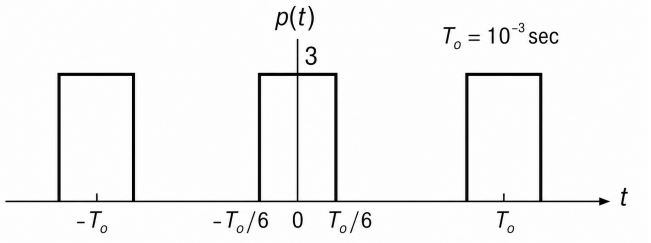


Fig. Q4.03.2

A. 2.7, 3.4 **B.** 3.3, 3.6
C. 2.6, 2.7 **D.** 2.7, 3.3

Q4.02.1 MCQ 2002 1M

The Fourier transform F $\{e^{-t}u(t)\}$ is equal to $\frac{1}{1+j2\pi f}$. Therefore, $F\left\{\frac{1}{1+j2\pi t}\right\}$ is

A. $e^{f}u(f)$ **B.** $e^{-f}u(f)$
C. $e^{f}u(-f)$ **D.** $e^{-f}u(-f)$

Q4.02.2 MCQ 2002 1M

Consider a sampled signal

$$y(t) = 5\times10^{-6}x(t)\sum_{n=-\infty}^{+\infty}\delta(t-nT_s)$$

where $x(t) = 10\cos(8\pi\times10^3)t$ and $T_s = 100\,\mu$sec. When $y(t)$ is passed through an ideal low-pass filter with a cutoff frequency of 5 KHz, the output of the filter is

A. $5\times10^{-6}\cos(8\pi\times10^3)t$

B. $5\times10^{-5}\cos(8\pi\times10^3)t$

C. $5\times10^{-1}\cos(8\pi\times10^3)t$

D. $10\cos(8\pi\times10^3)t$

Q4.02.3 MCQ 2002 2M

A signal $x(t) = 100\cos(24\pi\times10^3)t$ is ideally sampled with a sampling period of 50 μsec and then passed through an ideal low pass filter with cutoff frequency of 15 KHz. Which of the following frequencies is/ are present at the filter output?

A. 12 KHz only **B.** 8 KHz only
C. 12 KHz and 9 KHz **D.** 12 KHz and 8 KHz

Q4.01.1 MCQ 2001 2M

The Nyquist sampling interval, for the signal $\text{Sinc}(700t) + \text{Sinc}(500t)$ is

A. $\frac{1}{350}$ sec **B.** $\frac{\pi}{350}$ sec
C. $\frac{1}{700}$ sec **D.** $\frac{\pi}{350}$ sec

Q4.01.2 MCQ 2001 1M

A band limited signal is sampled at the Nyquist rate. The signal can be recovered by passing the samples through

A. an RC filter

B. an envelope detector

C. a PLL

D. an ideal low-pass filter with the appropriate bandwidth

Q4.00.1 MCQ 2000 1M

The Fourier Transform of the signal $x(t) = e^{-3t^2}$ is of the following form, where A and B are constants:

A. $Ae^{-B|f|}$ **B.** Ae^{-Bf}
C. $A + B|f|^2$ **D.** Ae^{-Bf^2}

Q4.00.2 MCQ 2000 2M

The Hilbert transform of $[\cos\omega_1 t + \sin\omega_2 t]$ is

A. $\sin\omega_1 t - \cos\omega_2 t$ **B.** $\sin\omega_1 t + \cos\omega_2 t$
C. $\cos\omega_1 t + \sin\omega_2 t$ **D.** $\sin\omega_1 t + \sin\omega_2 t$

Q4.99.1 MCQ 1999 1M

A signal $x(t)$ has a Fourier transform $X(\omega)$. If $x(t)$ is a real and odd function of t, then $X(\omega)$ is

A. a real and even function of ω

B. an imaginary and odd function of ω

C. an imaginary and even function of ω

D. a real and odd function of ω

Q4.99.2 MCQ 1999 2M

The Nyquist sampling frequency (in Hz) of a signal given by $16 \times 10^4 \sin c^2(400t) * 10^6 \sin c^3(100t)$ is

A. 200 **B.** 300
C. 500 **D.** 1000

Q4.98.1 MCQ 1998 1M

The Fourier transform of a voltage signal $x(t)$ is $X(f)$. The unit of $|X(f)|$ is

A. Volt **B.** Volt - sec
C. Volt/sec **D.** $Volt^2$

Q4.98.2 MCQ 1998 1M

The amplitude spectrum of a Gaussian pulse is

A. uniform **B.** a sine function
C. Gaussian **D.** an impulse function

Q4.98.3 MCQ 1998 1M

The Fourier transform of a function $x(t)$ is $X(f)$. The Fourier transform of $\frac{dx(t)}{dt}$ will be

A. $\frac{dX(f)}{dt}$ **B.** $j2\pi fX(f)$
C. $jfX(f)$ **D.** $\frac{X(f)}{jf}$

Q4.98.4 MCQ 1998 1M

Flat top sampling of low pass signals

A. gives rise to aperture effects

B. implies over sampling

C. leads to aliasing

D. introduces delay distortion

Q4.97.1 MCQ 1997 1M

The function $f(t)$ has the Fourier Transform $g(\omega)$. The Fourier Transform of

$$\mathcal{F}\{g(t)\} = \left(\int_{-\infty}^{\infty} g(t)e^{-j\omega t}\right) is$$

A. $\frac{1}{2\pi}f(\omega)$ **B.** $\frac{1}{2\pi}f(-\omega)$
C. $2\pi f(-\omega)$ **D.** none of the above

Q4.97.2 MCQ 1997 2M

If the Fourier Transfrom of a deterministic signal $g(t)$ is $G(f)$, then

Item-1

(1) The Fourier transform of $g(t-2)$ is

(2) The Fourier transform of $g(t/2)$ is

Item - 2

(A) $G(f)e^{-j(4\pi f)}$

(B) $G(2f)$

(C) $2G(2f)$
(D) $G(f-2)$

Match each of the items 1, 2 on the left with the most appropriate item A, B, C or D on the right.

A. (1 - A), (2 - C) **B.** (I - B), (2 - C)
C. (1 - D), (2 - C) **D.** (1 - A), (2 - D)

Q4.96.1 MCQ 1996 1M

The Fourier transform of a real valued time signal has

A. odd symmetry **B.** even symmetry
C. conjugate symmetry **D.** no symmetry

Q4.95.1 MCQ 1995 1M

A 1.0 kHz signal is flat - top sampled at the rate of 1800 samples/sec and the samples are applied to an ideal rectangular **LPF** with cut - off frequency of 1100 Hz, then the output of the filter contains

A. only 800 Hz component.

B. 800 Hz and 900 Hz components.

C. 800 Hz and 1000 Hz components.

D. 800 Hz, 900 Hz and 1000 Hz components.

Q4.94.1 MCQ 1994 1M

Increased pulse-width in the flat-top sampling, leads to

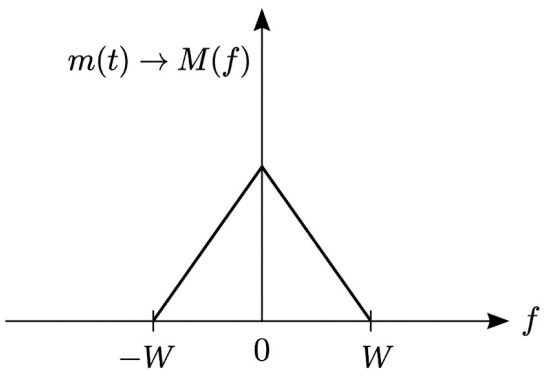


Fig. Q4.94.1

A. attenuation of high frequencies in reproduction

B. attenuation of low frequencies in reproduction

C. greater aliasing errors in reproduction

D. no harmful effects in reproduction

Q4.92.1 MCQ 1992 2M

If $G(f)$ represents the Fourier transform of a signal $g(t)$ which is real and odd symmetric in time, then

A. $G(f)$ is complex

B. $G(f)$ is imaginary

C. $G(f)$ is real

D. $G(f)$ is real and non- negative

Q4.91.1 NAT 1991 2M

A signal has frequency components from 300 Hz to 1.8 kHz. The minimum possible rate at which the signal has to be sampled is _____ kHz (fill in the blank).

Q4.88.1 MCQ 1988 2M

A signal containing only two frequency components (3 kHz and 6 kHz) is sampled at the rate of 8 kHz, and then passed through a low pass filter with a cutoff frequency of 8 kHz. The filter output

A. is an undistorted version of the original signal.

B. contains only the 3 kHz component.

C. contains the 3 kHz component and a spurious component of 2 kHz.

D. contains both the components of the original signal and two spurious components of 2 kHz and 5 kHz.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.26.1 D	Q4.26.2 B	Q4.25.1 A
Q4.25.2 A	Q4.24.1 *	Q4.24.2 B
Q4.23.1 C	Q4.23.2 15	Q4.23.3 B
Q4.23.4 A	Q4.22.1 A	Q4.21.1 B
Q4.20.1 58.64	Q4.18.1 8	Q4.18.2 13
Q4.17.1 B	Q4.17.2 8	Q4.17.3 8

Q4.16.1 13	Q4.16.2 A	Q4.16.3 0.25
Q4.16.4 B	Q4.16.5 A	Q4.16.6 C
Q4.15.1 0.4	Q4.15.2 A	Q4.15.3 3
Q4.14.1 D	Q4.14.2 0.2	Q4.14.3 B
Q4.14.4 A	Q4.14.5 3	Q4.13.1 A
Q4.12.1 C	Q4.10.1 C	Q4.08.1 A
Q4.08.2 D	Q4.08.3 C	Q4.06.1 A
Q4.06.2 C	Q4.06.3 C	Q4.05.1 B
Q4.05.2 A	Q4.04.1 B	Q4.04.2 C
Q4.04.3 C	Q4.03.1 C	Q4.03.2 D
Q4.02.1 C	Q4.02.2 C	Q4.02.3 D
Q4.01.1 C	Q4.01.2 D	Q4.00.1 D
Q4.00.2 A	Q4.99.1 B	Q4.99.2 B
Q4.98.1 B	Q4.98.2 C	Q4.98.3 B
Q4.98.4 A	Q4.97.1 C	Q4.97.2 A
Q4.96.1 C	Q4.95.1 C	Q4.94.1 A
Q4.92.1 B	Q4.91.1 3.6	Q4.88.1 D

UNIT

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 40	**MCQ:** 35	**MSQ:** 0	**NAT:** 5
1M: 13	**2M:** 27	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–3M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 1M

The Laplace Transform of the signal $x(t) = u(t-2) * (t\,u(t))$ is given by which of the following expressions? ["$*$" represents convolution operator]

A. $\frac{e^{-2s}}{s^2(s-2)}$ **B.** $\frac{e^{-2(s-2)}}{s^3}$

C. $\frac{s\,e^{-2s}}{(s-2)^2}$ **D.** $\frac{e^{-2s}}{s^3}$

Q5.16.1 MCQ 2016 2M

The Laplace transform of the causal periodic square wave of period T shown in the figure below is

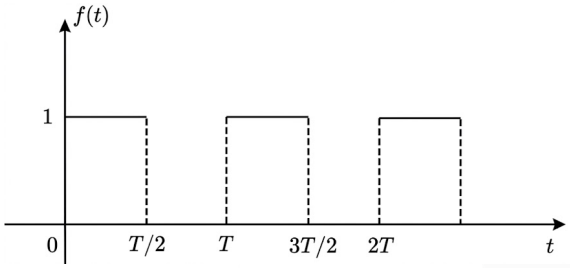


Fig. Q5.16.1

A. $F(s) = \frac{1}{1+e^{-sT/2}}$ **B.** $F(s) = \frac{1}{s(1+e^{-sT/2})}$

C. $F(s) = \frac{1}{s(1-e^{-sT})}$ **D.** $F(s) = \frac{1}{1-e^{-sT}}$

Q5.15.1 NAT 2015 2M

Let $x(t) = \alpha s(t) + s(-t)$ with $s(t) = \beta e^{-4t}u(t)$, where u(t) is unit step function. If the bilateral Laplace transform of x(t) is

$$X(s) = \frac{16}{s^2-16} \quad -4 < Re\{s\} < 4;$$

then the value of β is _____.

Q5.15.2 MCQ 2015 1M

The bilateral Laplace transform of a function

$$f(t) = \begin{cases} 1 & \text{if } a \le t \le b \\ 0 & \text{otherwise} \end{cases}$$

is

A. $\frac{a-b}{s}$ **B.** $\frac{e^{x}(a-b)}{s}$

C. $\frac{e^{-as}-e^{-bs}}{s}$ **D.** $\frac{e^{s(a-b)}}{s}$

Q5.15.3 MCQ 2015 2M

The solution of the differential equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$ with $y(0) = y'(0) = 1$ is

A. $(2-t)e^{t}$ **B.** $(1+2t)e^{-t}$

C. $(2+t)e^{-t}$ **D.** $(1-2t)e^{t}$

Q5.15.4 MCQ 2015 2M

The output of a standard second-order system for a unit step input is given as

$$y(t) = 1 - \frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t - \frac{\pi}{6}\right)$$

The transfer function of the system is

A. $\frac{2}{(s+2)(s+\sqrt{3})}$ **B.** $\frac{1}{s^2+2s+1}$

C. $\frac{3}{s^2+2s+3}$ **D.** $\frac{4}{s^2+2s+4}$

Q5.14.1 NAT 2014 2M

A causal LTI system has zero initial conditions and impulse response $h(t)$. Its input $x(t)$ and output $y(t)$ are related through the linear constant-coefficient differential equation

$$\frac{d^2y(t)}{dt^2} + \alpha\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t)$$

Let another signal $g(t)$ be defined as

$$g(t) = \alpha^2\int_0^t h(\tau)\,d\tau + \frac{dh(t)}{dt} + \alpha h(t)$$

If $G(s)$ is the Laplace transform of $g(t)$, then the number of poles of $G(s)$ is_____.

Q5.14.2 MCQ 2014 2M

A stable linear time invariant (LTI) system has a transfer function $H(S) = \frac{1}{s^2+s-6}$. To make this system causal it needs to be cascaded with another LTI system having a transfer function $H_1(S)$. A correct choice for $H_1(S)$ among the following options is

A. $s+3$ **B.** $s-2$

C. $s-6$ **D.** $s+1$

Q5.14.3 MCQ 2014 2M

The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. Which one of the following is the unilateral Laplace transform of $g(t) = t \cdot f(t)$?

A. $\frac{-s}{(s^2+s+1)^2}$ **B.** $\frac{-(2s+1)}{(s^2+s+1)^2}$

C. $\frac{s}{(s^2+s+1)^2}$ **D.** $\frac{2s+1}{(s^2+s+1)^2}$

Q5.14.4 MCQ 2014 2M

Let $h(t)$ denote the impulse response of a causal system with transfer function $\frac{1}{s+1}$. Consider the following three statements.

S1 : The system is stable

S2: $\frac{h(t+1)}{h(t)}$ is independent of $t > 0$

S3 : A non-causal system with the same transfer function is stable

For the above system

A. only S1 and S2 are true

B. only S2 and S3 are true

C. only S1 and S3 are true

D. S1,S2 and S3 are true

Q5.14.5 MCQ 2014 2M

A system is described by the following differential equation, where $u(t)$ is the input to the system and $y(t)$ is the output of the system.

$$\dot{y}(t) + 5y(t) = u(t)$$

When $y(0) = 1$ and $u(t)$ is a unit step function, $y(t)$ is

A. $0.2 + 0.8e^{-5t}$

B. $0.2 - 0.2e^{-5t}$

C. $0.8 + 0.2e^{-5t}$

D. $0.8 - 0.8e^{-5t}$

Q5.14.6 NAT 2014 1M

The input $-3e^{2t}u(t)$, where $u(t)$ is the unit step function, is applied to a system with transfer function $\frac{s-2}{s+3}$. If the initial value of the output is -2, then the value of the output at steady state is _____.

Q5.14.7 NAT 2014 1M

A continuous, linear time-invariant filter has an impulse response $h(t)$ described by

$$h(t) = \begin{cases} 3 & \text{for } 0 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$$

When a constant input of value 5 is applied to this filter, the steady state output is _____.

Q5.13.1 MCQ 2013 2M

A system is described by the differential equation

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = x(t)$$

Let $x(t)$ be a rectangular pulse given by

$$x(t) = \begin{cases} 1 & 0 < t < 2 \\ 0 & \text{otherwise} \end{cases}$$

Assuming that $y(0) = 0$ and $\frac{dy}{dt} = 0$ at $t = 0$, the Laplace transform of $y(t)$ is

A. $\frac{e^{-2s}}{s(s+2)(s+3)}$

B. $\frac{1-e^{-2s}}{s(s+2)(s+3)}$

C. $\frac{e^{-2s}}{(s+2)(s+3)}$

D. $\frac{1-e^{-2s}}{(s+2)(s+3)}$

Q5.11.1 MCQ 2011 2M

If $F(s) = L[f(t)] = \frac{2(s+1)}{s^2+4s+7}$ then the initial and final values of $f(t)$ are respectively

A. $0, 2$

B. $2, 0$

C. $0, 2/7$

D. $2/7, 0$

Q5.10.1 MCQ 2010 2M

Given $f(t) = l^{-1}\left[\frac{3s+1}{s^3+4s^2+(K-3)s}\right]$. If $\lim_{t\to\infty} f(t) = 1$, then the value of K is

A. 1

B. 2

C. 3

D. 4

Q5.09.1 MCQ 2009 2M

Given that $F(s)$ is the one-side Laplace transform of $f(t)$, the Laplace transform of $\int_0^t f(\tau)d\tau$ is

A. $sF(s) - f(0)$

B. $\frac{1}{s}F(s)$

C. $\int_0^s F(\tau)d\tau$

D. $\frac{1}{s}[F(s) - f(0)]$

Q5.07.1 MCQ 2007 1M

If the Laplace transform of a signal $y(t)$ is $Y(s) = \frac{1}{s(s-1)}$, then its final value is

A. -1

B. 0

C. 1

D. Unbounded

Q5.07.2 MCQ 2007 2M

The frequency response of a linear, time-invariant system is given by $H(f) = \frac{5}{1+j10\pi f}$. The step response of the system is:

A. $5(1 - e^{-5t})u(t)$

B. $5\left(1 - e^{-\frac{1}{5}t}\right)u(t)$

C. $\frac{1}{5}(1 - e^{-5t})u(t)$

D. $\frac{1}{5}\left(1 - e^{-\frac{1}{5}t}\right)u(t)$

Q5.06.1 MCQ 2006 2M

Consider the function $f(t)$ having Laplace transform $F(s) = \frac{\omega_0}{s^2+\omega_0^2}$ $Re[s] > 0$ The final value of $f(t)$ would be

A. 0

B. 1

C. $-1 \le f(\infty) \le 1$

D. ∞

Q5.06.2 MCQ 2006 2M

The unit-step response of a system starting from rest is given by $c(t) = 1 - e^{-2t}(t \ge 0)$, the transfer function of the system is

A. $\frac{1}{1+2s}$ **B.** $\frac{2}{2+s}$
C. $\frac{1}{2+s}$ **D.** $\frac{2s}{1+2s}$

Q5.06.3 MCQ 2006 2M

The unit impulse response of a system is $h(t) = e^{-t}, t \ge 0$. For this system, the steady-state value of the output for unit step input is equal to

A. -1 **B.** 0
C. 1 **D.** ∞

Q5.05.1 MCQ 2005 2M

In what range should $\mathrm{Re}(s)$ remain so that the Laplace transform of the function $e^{(a+2)t+5}$ exits.

A. $\mathrm{Re}(s) > a + 2$

B. $\mathrm{Re}(s) > a + 7$

C. $\mathrm{Re}(s) < 2$

D. $\mathrm{Re}(s) > a + 5$

Q5.04.1 MCQ 2004 2M

A system described by the following differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x(t)$$

is initially at rest. For input $x(t) = 2u(t)$, the output $y(t)$ is

A. $(1 - 2e^{-t} + e^{-2t})u(t)$

B. $(1 + 2e^{-t} - 2e^{-2t})u(t)$

C. $(0.5 + e^{-t} + 1.5e^{-2t})u(t)$

D. $(0.5 + 2e^{-t} + 2e^{-2t})u(t)$

Q5.03.1 MCQ 2003 1M

The Laplace transform of $i(t)$ is given by $I(s) = \frac{2}{s(1+s)}$ At $t \to \infty$, The value of $i(t)$ tends to

A. 0

B. 1

C. 2

D. ∞

Q5.02.1 MCQ 2002 2M

The Laplace transform of a continuous-time signal $x(t)$ is $X(s) = \frac{5-s}{s^2-s-2}$. If the Fourier transform of this signal exists, then $x(t)$ is

A. $e^{2t}u(t) - 2e^{-t}u(t)$

B. $-e^{2t}u(-t) + 2e^{-t}u(t)$

C. $-e^{2t}u(-t) - 2e^{-t}u(t)$

D. $e^{2t}u(-t) - 2e^{-t}u(t)$

Q5.00.1 MCQ 2000 1M

Given that

$$L[f(t)] = \frac{s+2}{s^2+1}, \qquad L[g(t)] = \frac{s^2+1}{(s+3)(s+2)}$$

and

$$h(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau.$$

Then, $L[h(t)]$ is

A. $\frac{s^2+1}{s+3}$ **B.** $\frac{1}{s+3}$

C. $\frac{s^2+1}{(s+3)(s+2)} + \frac{s+2}{s^2+1}$ **D.** None of the above

Q5.00.2 MCQ 2000 2M

A linear time invariant system has an impulse response e^{2t}, $t > 0$. If the initial conditions are zero and the input is e^{3t}, then output for $t > 0$ is

A. $e^{3t} - e^{2t}$ **B.** e^{5t}

C. $e^{3t} + e^{2t}$ **D.** None

Q5.99.1 MCQ 1999 1M

If $[f(t)] = F(s)$, then $[f(t-T)]$ is equal to

A. $e^{sT}F(s)$ **B.** $e^{-sT}F(s)$

C. $\frac{F(s)}{1+e^{sT}}$ **D.** $\frac{F(s)}{1-e^{-sT}}$

Q5.98.1 MCQ 1998 1M

If the Laplace transform of $f(t)$ is $F(s) = \frac{\omega}{s^2+\omega^2}$, then the value of $\lim_{t\to\infty} f(t)$

A. cannot be determined

B. is zero

C. is unity

D. is infinite

Q5.98.2 MCQ 1998 2M

The unit impulse response of a linear time invariant system is the unit step function $u(t)$. For $t > 0$, the response of the system ot an excitation $e^{-at}u(t)$, $a > 0$ will be

A. ae^{-at} **B.** $(1/a)(1-e^{-at})$

C. $a(1-e^{-at})$ **D.** $1-e^{-at}$

Q5.97.1 MCQ 1997 1M

The laplace transform of $e^{\alpha t}\cos\alpha t$ is equal to

A. $\frac{s-\alpha}{(s-\alpha)^2+\alpha^2}$

B. $\frac{s+\alpha}{(s+\alpha)^2+\alpha^2}$

C. $\frac{1}{(s-\alpha)^2}$

D. None

Q5.96.1 MCQ 1996 2M

The inverse Laplace transform of the function $\frac{s+5}{(s+1)(s+3)}$ is

A. $2e^{-t} - e^{-3t}$ **B.** $2e^{-t} + e^{-3t}$

C. $e^{-t} - 2e^{-3t}$ **D.** $e^{-t} + 2e^{-3t}$

Q5.95.1 MCQ 1995 1M

The final value theorem is used to find the

A. steady state value of the system output

B. initial value of the system output

C. transient behavior of the system output

D. none of these

Q5.95.2 MCQ 1995 1M

If $L[f(t)] = \frac{2(s+1)}{s^2+2s+5}$, then $f(0+)$ and $f(\infty)$ are given by

A. 0, 2 respectively **B.** 2, 0 respectively

C. 0, 1 respectively **D.** 2/5, 0 respectively

Q5.94.1 MCQ 1994 1M

The laplace transform of a unit ramp function starting at t=a, is

A. $\frac{1}{(s+a)^2}$

B. $\frac{e^{-as}}{(s+a)^2}$

C. $\frac{e^{-as}}{s^2}$

D. $\frac{a}{s^2}$

Q5.93.1 MCQ 1993 2M

If $F(s) = L[f(t)] = \frac{K}{(s+1)(s^2+4)}$ then $\lim_{t\to\infty} f(t)$ is given by

A. $K/4$

B. zero

C. infinite

D. undefined

Q5.93.2 NAT 1993 2M

The Laplace transform of the periodic function $f(t)$ described by the curve below, i.e.,

$$f(t) = \begin{cases} \sin t & if\ (2n-1)\pi \le t \le 2n\pi \\ 0 & otherwise \end{cases}$$

is _____. (fill in the blank), n is an integer.

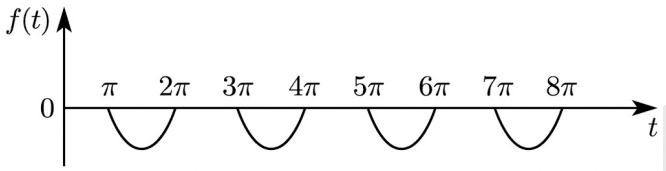


Fig. Q5.93.2

Q5.88.1 MCQ 1988 2M

The Laplace transform of a function $f(t)u(t)$, where $f(t)$ is periodic with period T, is $A(s)$ times the Laplace transform of its first period. Then

A. $A(s) = s$

B. $A(s) = 1/(1 - \exp(-Ts))$

C. $A(s) = 1/(1 + \exp(-Ts))$

D. $A(s) = \exp(Ts)$

Q5.87.1 MCQ 1987 2M

Laplace transform of the functions $tu(t)$ and $u(t)\sin(t)$ are respectively:

A. $\frac{1}{s^2}, \frac{s}{s^2+1}$

B. $\frac{1}{s}, \frac{1}{s^2+1}$

C. $\frac{1}{s^2}, \frac{1}{s^2+1}$

D. $s, \frac{s}{s^2+1}$

Answer Key

Q5.26.1 D	Q5.16.1 B	Q5.15.1 -2
Q5.15.2 C	Q5.15.3 B	Q5.15.4 D
Q5.14.1 1	Q5.14.2 B	Q5.14.3 D
Q5.14.4 A	Q5.14.5 A	Q5.14.6 -0.01 to 0.01
Q5.14.7 45	Q5.13.1 B	Q5.11.1 B
Q5.10.1 D	Q5.09.1 B	Q5.07.1 D
Q5.07.2 B	Q5.06.1 C	Q5.06.2 B
Q5.06.3 C	Q5.05.1 A	Q5.04.1 A
Q5.03.1 C	Q5.02.1 C	Q5.00.1 B
Q5.00.2 A	Q5.99.1 B	Q5.98.1 A
Q5.98.2 B	Q5.97.1 A	Q5.96.1 A
Q5.95.1 A	Q5.95.2 B	Q5.94.1 C
Q5.93.1 D	Q5.93.2 $-\frac{e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})}$	Q5.88.1 B
Q5.87.1 C		

UNIT

VI

Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 37	**MCQ:** 30	**MSQ:** 2	**NAT:** 5
1M: 22	**2M:** 15	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 1M

The response of a discrete time system $y[n]$ obeys the following relation: $y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$. The input to the system is $x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$. Which of the following options is TRUE for $y[n]$?

A. Stable and causal response

B. Stable and non-causal response

C. Unstable and causal response

D. Unstable and non-causal response

Q6.25.1 MSQ 2025 1M

Let $x[n]$ be a discrete-time signal whose z-transform is $X(z)$. Which of the following statements is/are TRUE?

☐ **A.** The discrete-time Fourier transform (DTFT) of $x[n]$ always exists

☐ **B.** The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros

☐ **C.** The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle

☐ **D.** If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the region of convergence (RoC) is the entire z-plane

Q6.24.1 MSQ 2024 1M

For a causal discrete-time LTI system with transfer function $H(z) = \frac{2z^2+3}{(z+\frac{1}{3})(z-\frac{1}{3})}$ which of the following statements is/are true?

☐ **A.** The system is stable.

☐ **B.** The system is a minimum phase system.

☐ **C.** The initial value of the impulse response is 2.

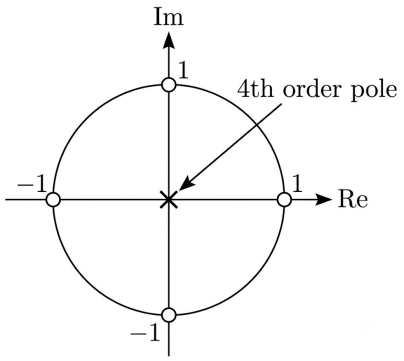
☐ **D.** The final value of the impulse response is 0.

Q6.20.1 MCQ 2020 1M

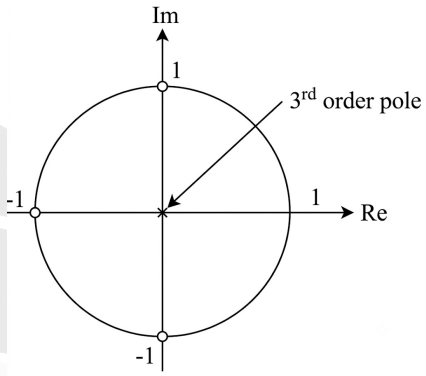
Which one of the following pole-zero corresponds to the transfer function of an LTI system characterized by the input-output difference equation given below?

$$y[n] = \sum_{k=0}^{3}(-1)^k x[n-k]$$

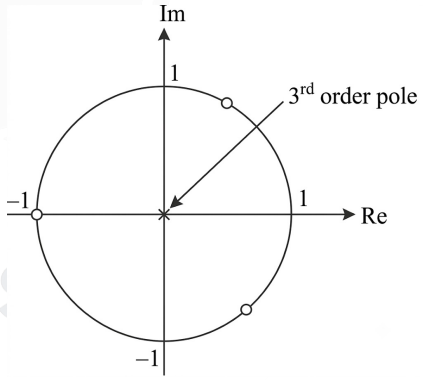
A.



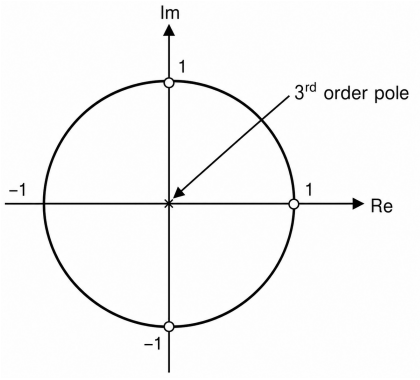
B.



C.



D.



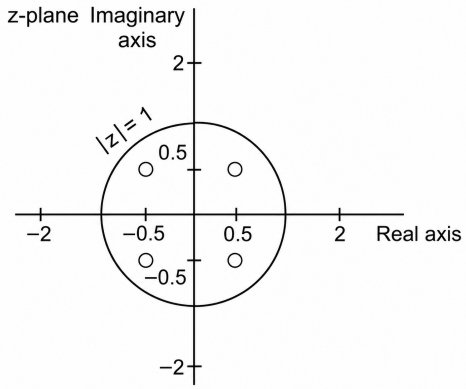
Q6.20.2 NAT 2020 2M

The transfer function of a stable discrete-time LTI system is $H(z) = \frac{K(z-\alpha)}{z+0.5}$, where K and α are real numbers. The value of α (rounded off to one decimal place) with $|\alpha| > 1$, for which the magnitude response of the system is constant over all frequencies, is _____.

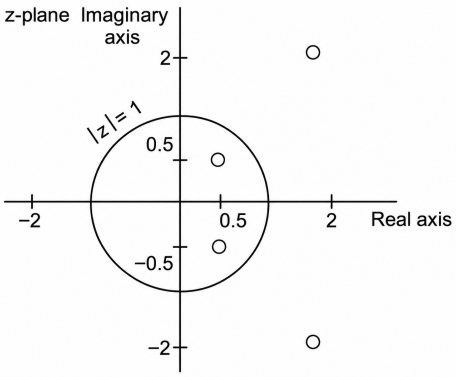
Q6.19.1 MCQ 2019 1M

Let $H(z)$ be the z-transform of a real-valued discrete-time signal $h[n]$. If $P(z) = H(z)H\left(\frac{1}{z}\right)$ has a zero at $z = \frac{1}{2} + \frac{1}{2}j$, and $P(z)$ has a total of four zeros, which one of the following plots represents all the zeros correctly?

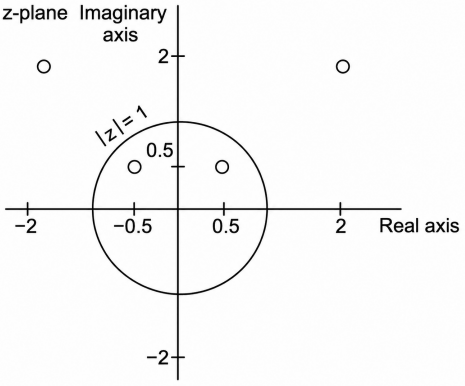
A.



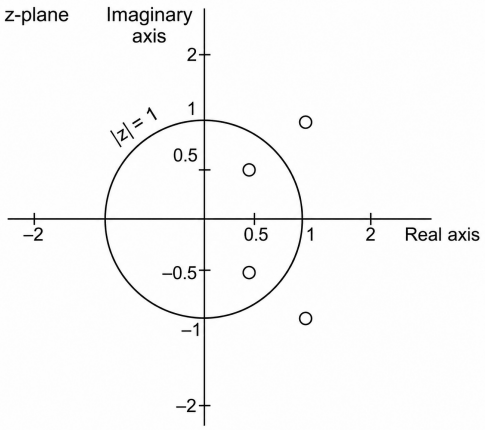
B.



C.



D.



Q6.18.1 MCQ 2018 1M

A discrete time all-pass system has two of its poles at $0.25\angle 0°$ and $2\angle 30°$. Which one of the following statements about the system is TRUE?

A. It has two more poles at $0.5\angle 30°$ and $4\angle 0°$.

B. It is stable only when the impulse response is two-sided.

C. It has constant phase response over all frequencies.

D. It has constant phase response over the entire z-plane.

Q6.16.1 MCQ 2016 1M

Consider the sequence $x[n] = a^n u[n] + b^n u[n]$, where u[n] denotes the unit step sequence and $0 < |a| < |b| < 1$. The region of convergence (ROC) of the z-transform of $[n]$ is

A. $|z| > |a|$

B. $|z| > |b|$

C. $|z| < |a|$

D. $|a| < |z| < |b|$

Q6.16.2 MCQ 2016 1M

A discrete-time signal $x[n] = \delta[n-3] + 2\delta[n-5]$ has z-transform $x(z)$. If $Y(z) = X(-z)$ is the z-transform of another signal $y[n]$, then

A. $y[n] = x[n]$

B. $y[n] = x[-n]$

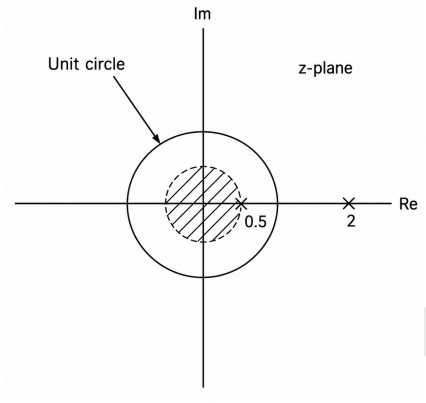
C. $y[n] = -x[n]$

D. $y[n] = -x[-n]$

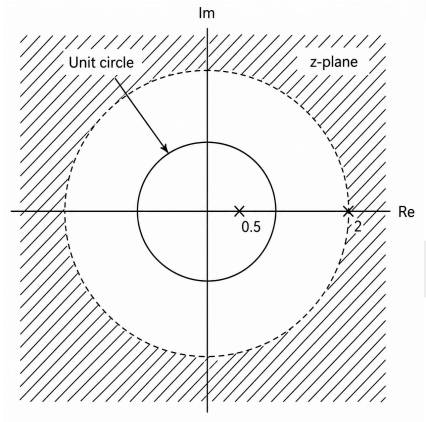
Q6.16.3 MCQ 2016 2M

The ROC (region of convergence) of the z-transform of a discrete-time signal is represented by the shaded region in the z-plane. If the signal $x[n] = (2.0)^{|n|}$, $-\infty < n < +\infty$, then the ROC of its z-transform is represented by

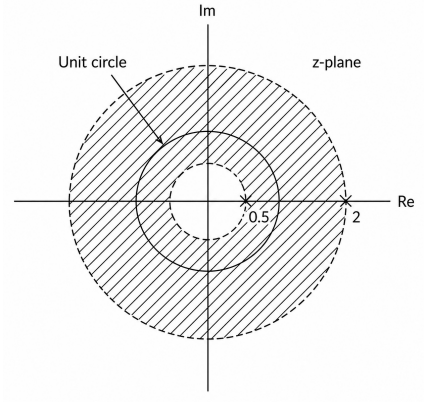
A.



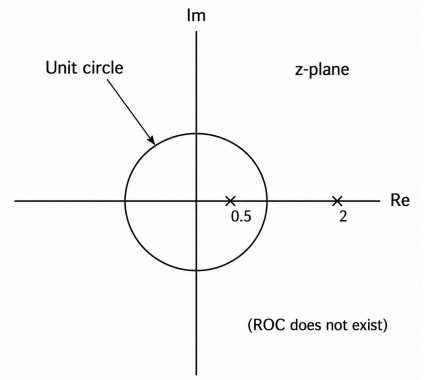
B.



C.



D.



Q6.15.1 MCQ 2015 2M

A realization of a stable discrete time system is shown in the figure. If the system is excited by a unit step sequence input $x[n]$, the response $y[n]$ is

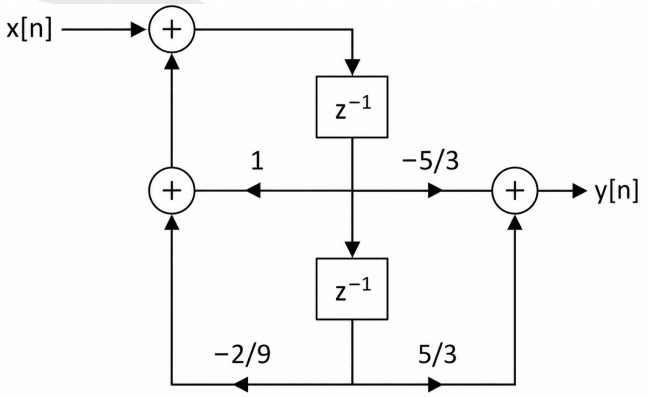


Fig. Q6.15.1

A. $4\left(-\frac{1}{3}\right)^n u[n] - 5\left(-\frac{2}{3}\right)^n u[n]$

B. $5\left(-\frac{2}{3}\right)^n u[n] - 3\left(-\frac{1}{3}\right)^n u[n]$

C. $5\left(\frac{1}{3}\right)^n u[n] - 5\left(\frac{2}{3}\right)^n u[n]$

D. $5\left(\frac{2}{3}\right)^n u[n] - 5\left(\frac{1}{3}\right)^n u[n]$

Q6.15.2 MCQ 2015 2M

Suppose $x[n]$ is an absolutely summable discrete-time signal. Its z-transform is a rational function with two poles and two zeroes. The poles are at $z = \pm 2j$. Which one of the following statements is TRUE for the signal $x[n]$?

A. It is a finite duration signal.

B. It is a causal signal

C. It is a non-causal signal

D. It is a periodic signal

Q6.15.3 MCQ 2015 2M

The pole-zero diagram of a causal and stable discrete-time system is shown in the figure. The zero at the origin has multiplicity 4. The impulse response of the system is $h[n]$. If $h[0] = 1$, we can conclude

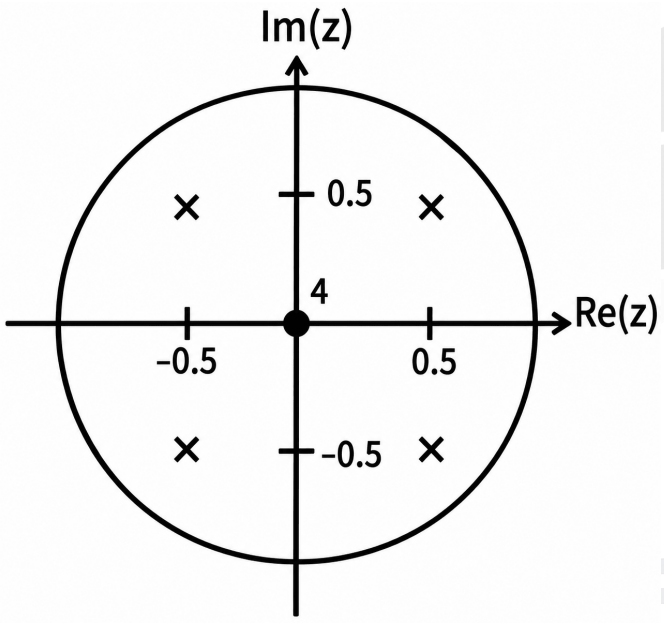


Fig. Q6.15.3

A. $h[n]$ is real for all n

B. $h[n]$ is purely imaginary for all n

C. $h[n]$ is real for only even n

D. $h[n]$ is purely imaginary for only odd n

Q6.15.4 MCQ 2015 2M

For the discrete-time system shown in the figure, the poles of the system transfer function are located at

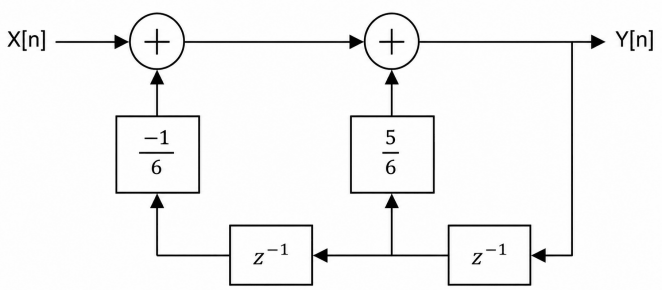


Fig. Q6.15.4

A. $2, 3$

B. $1/2, 3$

C. $1/2, 1/3$

D. $2, 1/3$

Q6.15.5 NAT 2015 1M

Two causal discrete-time signals $x[n]$ and $y[n]$ are re lated as $y[n] = \sum_{m=0}^{n} x[m]$. If the z-transform of $y[n]$ is $\frac{2}{z(z-1)^2}$, the value of $x[2]$ is _____.

Q6.14.1 NAT 2014 2M

The z-transform of the sequence $x[n]$ is given by $X(z) = \frac{1}{(1-2z^{-1})^2}$, with the region of convergence $|z| > 2$. Then, $x[2]$ is _____.

Q6.14.2 NAT 2014 2M

Let $H_1(z) = (1 - pz^{-1})^{-1}$, $H_2(z) = (1 - qz^{-1})^{-1}$, $H(z) = H_1(z) + rH_2(z)$. The quantities p, q, r are real numbers. Consider $p = \frac{1}{2}$, $q = -\frac{1}{4}$, $|r| < 1$. If the zero of $H(z)$ lies on the unit circle, then $r =$ _____.

Q6.14.3 MCQ 2014 2M

The input-output relationship of a causal stable LTI system is given as $y[n] = \alpha y[n-1] + \beta x[n]$ If the impulse response $h[n]$ of this system satisfies the con dition $\sum_{n=0}^{\infty} h[n] = 2$, the relationship between α and β is

A. $\alpha = 1 - \beta/2$

B. $\alpha = 1 + \beta/2$

C. $\alpha = 2\beta$

D. $\alpha = -2\beta$

Q6.14.4 MCQ 2014 2M

Let $x[n] = \left(-\frac{1}{9}\right)^n u(n) - \left(-\frac{1}{3}\right)^n u(-n-1)$. The Region of Convergence (ROC) of the z -transform of x[n].

A. is $|z| > \frac{1}{9}$

B. is $|z| < \frac{1}{3}$

C. is $\frac{1}{3} > |z| > \frac{1}{9}$

D. does not exist

Q6.14.5 MCQ 2014 1M

Let $x[n] = x[-n]$. Let $X(z)$ be the z -transform of $x[n]$. If $0.5 + j0.25$ is a zero of $X(z)$, which one of the following must also be a zero of $X(z)$.

A. $0.5 - j0.25$

B. $1/(0.5 + j0.25)$

C. $1/(0.5 - j0.25)$

D. $2 + j4$

Q6.14.6 MCQ 2014 1M

For an all-pass system $H(z) = \frac{(z^{-1}-b)}{(1-az^{-1})}$, where $|H(e^{-j\omega})| = 1$, for all ω. If $\text{Re}(a) \neq 0$, $\text{Im}(a) \neq 0$, then b equals

A. a

B. a^*

C. $1/a^*$

D. $1/a$

Q6.14.7 MCQ 2014 1M

An FIR system is described by the system function

$$H(z) = 1 + \frac{7}{2}z^{-1} + \frac{3}{2}z^{-2}$$

A. maximum phase

B. minimum phase

C. mixed phase

D. zero phase

Q6.14.8 NAT 2014 1M

The sequence $x[n] = 0.5^n u[n]$, where $u[n]$ is the unit step sequence, is convolved with itself to obtain $y[n]$. Then $\sum_{n=-\infty}^{+\infty} y[n]$ is _____.

Q6.12.1 MCQ 2012 1M

If $x[n] = (1/3)^{|n|} - (1/2)^n u[n]$, then the region of convergence (ROC) of its Z-transform in the Z-plane will be

A. $\frac{1}{3} < |z| < 3$ **B.** $\frac{1}{3} < |z| < \frac{1}{2}$
C. $\frac{1}{2} < |z| < 3$ **D.** $\frac{1}{3} < |z|$

Q6.10.1 MCQ 2010 1M

Consider the z-transform $X(z) = 5z^2 + 4z^{-1} + 3; 0 < |z| < \infty$. The inverse z - transform $x[n]$ is

A. $5\delta[n+2] + 3\delta[n] + 4\delta[n-1]$

B. $5\delta[n-2] + 3\delta[n] + 4\delta[n+1]$

C. $5u[n+2] + 3u[n] + 4u[n-1]$

D. $5u[n-2] + 3u[n] + 4u[n+1]$

Q6.09.1 MCQ 2009 1M

The ROC of Z-transform of the discrete time sequence $x(n) = \left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$ is

A. $\frac{1}{3} < |z| < \frac{1}{2}$ **B.** $|z| > \frac{1}{2}$
C. $|z| < \frac{1}{2}$ **D.** $2 < |z| < 3$

Common Data Questions

The next 2 questions share the data below.

In the following network, the switch is closed at $t = 0^-$ and the sampling starts from $t = 0$. The sampling frequency is 10 Hz.

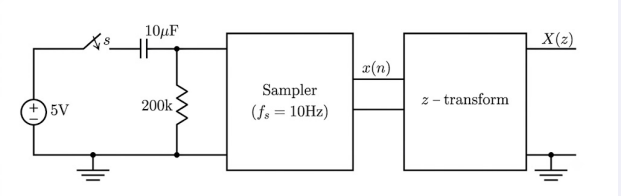


Fig. Q6.09.1

Q6.08.1 MCQ 2008 2M

The samples $x(n)$ $(n = 0, 1, 2, \ldots)$ are given by

A. $5\left(1 - e^{-0.05n}\right)$

B. $5e^{-0.05n}$

C. $5\left(1 - e^{-5n}\right)$

D. $5e^{-5n}$

Q6.08.2 MCQ 2008 2M

The expression and the region of convergence of the z-transform of the sampled signal are

A. $\frac{5z}{z - e^{-5}}, |z| < e^{-5}$ **B.** $\frac{5z}{z - e^{-0.05}}, |z| < e^{-0.05}$

C. $\frac{5z}{z - e^{-0.05}}, |z| > e^{-0.05}$ **D.** $\frac{5z}{z - e^{-5}}, |z| > e^{-5}$

Q6.07.1 MCQ 2007 2M

The z-transform $X(z)$ of a sequence $x[n]$ is given by $= \frac{0.5}{1-2z^{-1}}$. It is given that the region of convergence of $X[z]$ includes the unit circle. The value of $x[0]$ is

A. -0.5 **B.** 0

C. 0.25 **D.** 0.5

Q6.06.1 MCQ 2006 1M

If the region of convergence of $x_1[n] + x_2[n]$ is $1/3 < |z| < 2/3$, then the region of convergence of $x_1[n] - x_2[n]$ includes

A. $\frac{1}{3} < |z| < 3$ **B.** $\frac{2}{3} < |z| < 3$

C. $\frac{3}{2} < |z| < 3$ **D.** $\frac{1}{3} < |z| < \frac{2}{3}$

Q6.05.1 MCQ 2005 1M

The region of convergence of z-transform of the sequence $\left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$ must be

A. $|z| < \frac{5}{6}$ **B.** $|z| > \frac{6}{5}$

C. $\frac{5}{6} < |z| < \frac{6}{5}$ **D.** $\frac{6}{5} < |z| < \infty$

Q6.04.1 MCQ 2004 1M

The z transform of a system is

$$H(z) = \frac{z}{z - 0.2}.$$

If the ROC is $|z| < 0.2$, then the impulse response of the system is

A. $(0.2)^n u[n]$ **B.** $(0.2)^n u[-n-1]$

C. $-(0.2)^n u[n]$ **D.** $-(0.2)^n u[-n-1]$

Q6.01.1 MCQ 2001 1M

The region of convergence of the z- transform of a unit step function is

A. $|z| > 1$ **B.** $|z| < 1$

C. (Real part of z) > 0 **D.** (Real part of z) < 0

Q6.99.1 MCQ 1999 1M

The z-transform $F(z)$ of the function $f(nT) = a^{nT}$ is

A. $\frac{z}{z-a^T}$ **B.** $\frac{z}{z+a^T}$

C. $\frac{z}{z-a^{-T}}$ **D.** $\frac{z}{z+a^{-T}}$

Q6.99.2 MCQ 1999 2M

The z-transform of a signal is given by $c(z) = \frac{1}{4}\frac{z^{-1}(1-z^{-4})}{(1-z^{-1})^2}$. Its final value is

A. 1/4 **B.** zero

C. 1.0 **D.** infinity

Q6.98.1 MCQ 1998 1M

The z - transform of the time function $\sum_{k=0}^{\infty} \delta(n-k)$ is

A. $(z-1)/z$ **B.** $z/(z-1)^2$

C. $z/(z-1)$ **D.** $(z-1)^2/z$

Q6.90.1 MCQ 1990 2M

The Z-transform of the following real exponential sequence: $x(nT) = a^n, nT \geq 0 = 0, nT < 0, a > 0$ gives us by

A. $\frac{1}{1-z^{-1}}; |z| > 1$ **B.** $\frac{1}{1-az^{-1}}; |z| > a$

C. 1 for all z **D.** $\frac{1}{1-az^{-1}}; |z| < a$

Answer Key

Q6.26.1 A	**Q6.25.1** C,D	**Q6.24.1** A,C,D
Q6.20.1 D	**Q6.20.2** -2	**Q6.19.1** D
Q6.18.1 B	**Q6.16.1** B	**Q6.16.2** C
Q6.16.3 D	**Q6.15.1** C	**Q6.15.2** C
Q6.15.3 A	**Q6.15.4** C	**Q6.15.5** 0
Q6.14.1 12	**Q6.14.2** -0.5	**Q6.14.3** A

Q6.14.4 C	Q6.14.5 B	Q6.14.6 B
Q6.14.7 C	Q6.14.8 4	Q6.12.1 C
Q6.10.1 A	Q6.09.1 A	Q6.08.1 B
Q6.08.2 C	Q6.07.1 B	Q6.06.1 D
Q6.05.1 C	Q6.04.1 D	Q6.01.1 A
Q6.99.1 A	Q6.99.2 C	Q6.98.1 C
Q6.90.1 B		

PrepFusion

UNIT

VII

DTFT, DTFS, DFT & Miscellaneous

Topics

1. DTFT
2. Discrete-Time Fourier Series (DTFS)
3. Circular Convolution
4. DFT
5. Group Delay, Phase Delay and Basics of FFT

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 29	**MCQ:** 15	**MSQ:** 0	**NAT:** 14
1M: 5	**2M:** 24	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2014)–4M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q7.24.1 NAT 2024 2M

The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined as

$$X[k] = \mathcal{F}\{x[n]\}.$$

Another sequence $y[n]$ is formed as below

$$y[n] = \mathcal{F}\{\mathcal{F}\{\mathcal{F}\{\mathcal{F}\{x[n]\}\}\}\}.$$

For the sequence $x[n] = \{1, 2, 1, 3\}$, the value of $Y[0]$ is _____.

Q7.23.1 MCQ 2023 2M

Let an input $x[n]$ having discrete time Fourier transform $x(e^{j\Omega}) = 1 - e^{-j\Omega} + 2e^{-3j\Omega}$ be passed through an LTI system. The frequency response of the LTI system is $H(e^{j\Omega}) = 1 - \frac{1}{2}e^{-j2\Omega}$. The output $y[n]$ of the system is

A. $\delta[n] + \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$

B. $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$

C. $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$

D. $\delta[n] + \delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] + \delta[n-5]$

Q7.23.2 MCQ 2023 2M

Consider a discrete-time periodic signal with period $N = 5$. Let the discrete-time Fourier series (DTFS) representation be $x[n] = \sum_{k=0}^{4} a_k e^{\frac{jk2\pi n}{5}}$, where $a_0 = 1$, $a_1 = 3j$, $a_2 = 2j$, $a_3 = -2j$ and $a_4 = -3j$. The value of the sum $\sum_{n=0}^{4} x[n] \sin\frac{4\pi n}{5}$ is

A. -10 **B.** 10
C. -2 **D.** 2

Q7.22.1 NAT 2022 2M

For a vector $\overline{x} = [x[0], x[1], \ldots, x[7]]$, the 8-point discrete Fourier transform (DFT) is denoted by $\overline{X} = \text{DFT}(\overline{x}) = [X[0], X[1], \ldots, X[7]]$, where

$$X[k] = \sum_{n=0}^{7} x[n] \exp\left(-j\frac{2\pi}{8}nk\right).$$

Here, $j = \sqrt{-1}$. If $\overline{x} = [1, 0, 0, 0, 2, 0, 0, 0]$ and $\overline{y} = \text{DFT}(\text{DFT}(\overline{x}))$, then the value of $y[0]$ is _____ (rounded off to one decimal place).

Q7.21.1 MCQ 2021 1M

Consider two 16-point sequences $x[n]$ and $h[n]$. Let the linear convolution of $x[n]$ and $h[n]$ be denoted by $y[n]$, while $z[n]$ denotes the 16-point inverse discrete Fourier transform (IDFT) of the product of the 16-point DFTs of $x[n]$ and $h[n]$. The value(s) of k for which $z[k] = y[k]$ is/are :

A. $k = 0, 1, 2, \ldots, 15$ **B.** $k = 0$ and $k = 15$
C. $k = 0$ **D.** $k = 15$

Q7.21.2 NAT 2021 2M

Consider the signals $x[n] = 2^{n-1}u[-n+2]$ and $y[n] = 2^{-n+2}u[n+1]$, where $u[n]$ is the unit step sequence. Let $X(e^{j\omega})$ and $Y(e^{j\omega})$ be the discrete-time Fourier transform of $x[n]$ and $y[n]$, respectively. The value of the integral

$$\frac{1}{2\pi}\int_0^{2\pi} X(e^{j\omega})Y(e^{-j\omega})d\omega$$

(round off to one decimal place) is _____

Q7.20.1 MCQ 2020 2M

A finite duration discrete-time signal $x[n]$ is obtained by sampling a continuous - time signal $x(t) = \cos(200\pi t)$ at sampling instants $t = \frac{n}{400}, n = 0, 1, \ldots, 7$. The 8-point discrete Fourier transform (DFT) is defined a $X[k] = \sum_{n=0}^{7} x[n]e^{-j\pi nk/4}$ for $k = 0, 1, \ldots, 7$ Which one of the following statements is TRUE?

A. Only $X(3)$ and $X(5)$ are non-zero.

B. Only $X(4)$ is non-zero.

C. All $X(K)$ are non-zero.

D. Only $X(2)$ and $X(6)$ are non-zero.

Q7.18.1 NAT 2018 2M

Let $X[k] = k + 1, 0 \le k \le 7$ be 8-point DFT of a sequence $x[n]$, where $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi nk/N}$. The value (correct to two decimal places) of $\sum_{n=0}^{3} x[2n]$ is _____.

Q7.17.1 NAT 2017 2M

Two discrete-time signals $x[n]$ and $h[n]$ are both non-zero only for $n = 0, 1, 2$ and are zero otherwise. It is given that $x(0) = 1, x[1] = 2, x[2] = 1, h[0] = 1$, let $y[n]$ be the linear convolution of $x[n]$ and $h[n]$. Given that $y[1] = 3$ and $y[2] = 4$, the value of the expression $(10y[3] + y[4])$ is _____.

Q7.17.2 NAT 2017 2M

Let $h[n]$ be the impulse response of a discrete time linear time invariant (LTI) filter. The impulse response is given by $h(0) = \frac{1}{3}; h[1] = \frac{1}{3}; h[2] = \frac{1}{3}$; and $h[n] = 0$ for $n < 0$ and $n > 2$. Let $H(\omega)$ be the Discrete-time Fourier transform (DTFT) of $h[n]$, where ω is the normalized angular frequency in radians. Given that $H(\omega_o) = 0$ and $0 < \omega_0 < \pi$, the value of ω_o (in radians) is equal to _____.

Q7.16.1 NAT 2016 2M

The Discrete Fourier Transform (DFT) of the 4-point sequence $x[n] = \{x[0], x[1], x[2], x[3]\} = \{3, 2, 3, 4\}$ is $x[k] = \{X[0], X[1], X[2], X[3]\} = \{12, 2j, 0, -2j\}$ If $X_1[k]$ is the DFT of the 12- point sequence $x_1[n] = \{3, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0\}$, The value of $\left|\frac{X_1[8]}{X_1[11]}\right|$ is _____.

Q7.16.2 NAT 2016 2M

A continuous-time speech signal $x_a(t)$ is sampled at a rate of 8 kHz and the samples are subsequently grouped in blocks, each of size N. The DFT of each block is to be computed in real time using the radix-2 decimation-in-frequency FFT algorithm. If the processor performs all operations sequentially, and takes 20 μs for computing each complex multiplication (including multiplications by 1 and -1) and the time required for addition/ subtraction is negligible, then the maximum value of N is _____.

Q7.16.3 NAT 2016 2M

Consider the signal $x[n] = 6\delta[n+2] + 3\delta[n+1] + 8\delta[n] + 7\delta[n-1] + 4\delta[n-2]$. If $X(e^{t\omega})$is the discrete-time Fourier transform of $x[n]$, then

$$\frac{1}{\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2\omega)d\omega$$

is equal to _____.

Q7.15.1 NAT 2015 2M

Consider two real sequences with time- origin marked by the bold value, $x_1[n] = \{\mathbf{1}, 2, 3, 0\}, x_2[n] = \{\mathbf{1}, 3, 2, 1\}$ Let $X_1(k)$ and $X_2(k)$ be 4-point DFTs of $x_1[n]$ and $x_2[n]$, respectively. Another sequence $X_3(n)$ is derived by taking 4-ponit inverse DFT of $X_3(k) = X_1(k)X_2(k)$. The value of $x_3[2]$

Q7.15.2 MCQ 2015 2M

Two sequences $[a, b, c]$ and $[A, B, C]$ are related as,

$$\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{-1} & W_3^{-2} \\ 1 & W_3^{-2} & W_3^{-4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \text{Where}$$

$W_3 = e^{j\frac{2\pi}{3}}$. if another sequence $[p, q, r]$ is derived as,

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^1 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & W_3^2 & 0 \\ 0 & 0 & W_3^4 \end{bmatrix} \begin{bmatrix} A/3 \\ B/3 \\ C/3 \end{bmatrix},$$

Then the relationship between the sequences $[p, q, r]$ and $[a, b, c]$ is

A. $[p, q, r] = [b, a, c]$ **B.** $[p, q, r] = [b, c, a]$
C. $[p, q, r] = [c, a, b]$ **D.** $[p, q, r] = [c, b, a]$

Q7.15.3 NAT 2015 1M

The value of $\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n$ is _____.

Q7.15.4 NAT 2015 2M

Let $\widetilde{x}[n] = 1 + \cos\left[\frac{\pi n}{8}\right]$ be a periodic signal with period 16. Its DFS coefficients are defined by $a_k = \frac{1}{16}\sum_{n=0}^{15}\widetilde{x}[n]\exp\left(-j\frac{\pi}{8}kn\right)$ for all k. The value of the coeffcients a_{31} is _____.

Q7.14.1 NAT 2014 2M

Consider a discrete time periodic signal $x[n] = \sin\left(\frac{\pi n}{5}\right)$. Let a_k be the complex Fourier serier coefficients of $x[n]$. The coefficients $\{a_k\}$ are non- zero when $k = Bm \pm 1$, where m is any integer. The value of B is _____.

Q7.14.2 MCQ 2014 2M

The N-point DFT X of a sequence $x[n]$ $0 \le n \le N-1$ is given by

$$X[k] = \frac{1}{\sqrt{N}}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}nk}, 0 \le k \le N-1$$

Denote this relation as $X = \text{DFT}(x)$. For $N = 4$ which one of the following sequences satisfies DFT (DFT(x)) = x ?

A. x = [1 2 3 4] **B.** x = [1 2 3 2]
C. x = [1 3 2 2] **D.** x = [1 2 2 3]

Q7.14.3 NAT 2014 1M

A Fourier transform pair is given by

$$\left(\frac{2}{3}\right)^n u[n+3] \xleftrightarrow{FT} \frac{Ae^{j6\pi f}}{1-\left(\frac{2}{3}\right)e^{-j2\pi f}}$$

where $u(n)$ denotes the unit step sequence. The values of A is _____.

Q7.13.1 MCQ 2013 2M

The DFT of a vector $[a\ b\ c\ d]$ is the vector $[\alpha\ \beta\ \gamma\ \delta]$. Consider the product

$$[p\quad q\quad r\quad s] = [a\quad b\quad c\quad d]\begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}.$$

The DFT of the vector $[p\ q\ r\ s]$ is a scaled version of

A. $[\alpha^2\ \beta^2\ \gamma^2\ \delta^2]$
B. $\sqrt{\alpha}\ \sqrt{\beta}\ \sqrt{\gamma}\ \sqrt{\delta}$
C. $[\alpha+\beta\ \beta+\delta\ +\gamma\ \gamma+\alpha]$
D. $[\delta\ \beta\ \gamma\ \delta]$

Q7.11.1 MCQ 2011 2M

The first six points of the 8-point DFT of a real valued sequence are $5, 1-j3, 0, 3-j4, 0$ and $3+j4$. The last two points of the DFT are respectively

A. $0, 1-j3$
B. $0, 1+j3$
C. $1+j3, 5$
D. $1-j3, 5$

Q7.10.1 MCQ 2010 1M

For an N-point FFT algorithm with $N = 2^m$ which one of the following statements is TRUE?

A. It is not possible to construct a signal flow graph with both input and output in normal order.

B. The number of butterflies in the mth stage is N/m.

C. In-place computation requires storage of only $2N$ node data.

D. Computation of a butterfly requires only one complex multiplication.

Q7.09.1 MCQ 2009 2M

The 4-point Discrete Fourier Transform (DFT) of a discrete time sequence $\{1, 0, 2, 3\}$ is

A. $[0, -2+2j, 2, -2-2j]$
B. $[2, 2+2j, 6, 2-2j]$
C. $[6, 1-3j, 2, 1+3j]$
D. $[6, -1+3j, 0, -1-3j]$

Q7.08.1 MCQ 2008 2M

$\{x(n)\}$ is a real-valued periodic sequence with a period N. $x(n)$ and $X(k)$ form N-point. Discrete Fourier Transform (DFT) pairs. The DFT $Y(k)$ of the sequence $y(n) = \frac{1}{N}\sum_{r=0}^{N-1} x(r)x(n+r)$ is

A. $|X(k)|^2$
B. $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X^*(k+r)$
C. $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X(k+r)$
D. 0

Q7.07.1 MCQ 2007 2M

A 5-point sequence $x[n]$ is given as $x[-3] = 1, x[-2] = 1, x[-1] = 0, x[0] = 5, x[-1] = 1$. Let $X(e^{j\omega})$ denote the discrete - time Fourier transform of $x(n)$. The value of $\int_{-\pi}^{\pi} X((e^{j\omega})\, d\omega$ is

A. 5
B. 10π
C. 16π
D. $5 + j10\pi$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q7.05.1 & Q7.05.2: A sequence $x(n)$ has non-zero values as shown in figure.

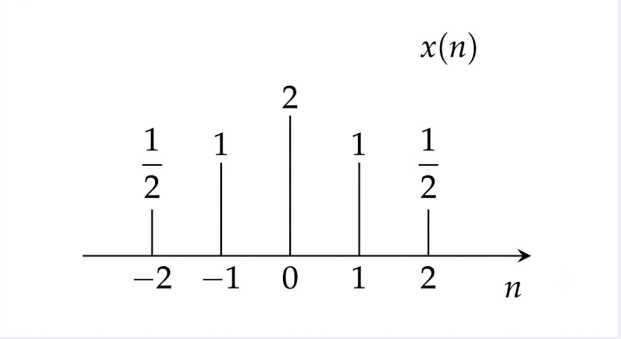


Fig. Q7.07.1

Q7.05.1 MCQ 2005 2M

The sequence

$$y(n) = \begin{cases} x\left(\frac{n}{2}-1\right) & \text{for } n \text{ even} \\ 0 & \text{for } n \text{ odd} \end{cases}$$

will be

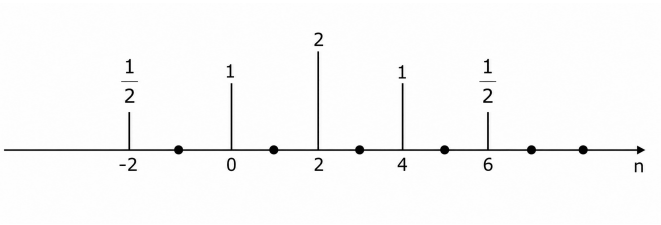


Fig. Q7.05.1

A.

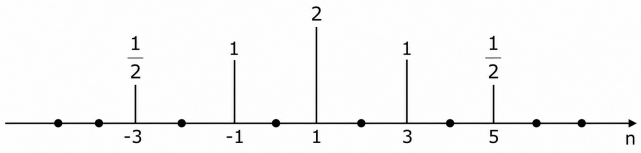


Fig. Q7.05.1

B.

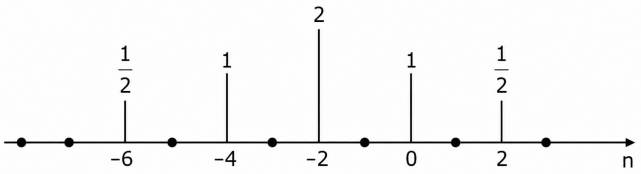


Fig. Q7.05.1

C.

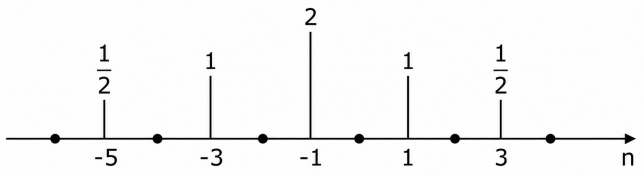


Fig. Q7.05.1

D.

Q7.05.2 MCQ 2005 2M

The Fourier transform of $y(2n)$ will be

A. $e^{-j2\omega}[\cos 4\omega + 2\cos 2\omega + 2]$ **B.** $[\cos 2\omega + 2\cos \omega + 2]$

C. $e^{-j\omega}[\cos 2\omega + 2\cos \omega + 2]$ **D.** $e^{j2\omega}[\cos 2\omega + 2\cos \omega + 2]$

Q7.05.3 MCQ 2005 1M

Let $x(n) = \left(\frac{1}{2}\right)^n u(n)$, $y(n) = x^2$, and $Y((e^{j\omega})$ be the Fourier transform of $y(n)$. Then $Y((e^{j0})$ is

A. $\frac{1}{4}$ **B.** 2

C. 4 **D.** $\frac{4}{3}$

Answer Key

Q7.24.1 112	Q7.23.1 C	Q7.23.2 A
Q7.22.1 8	Q7.21.1 D	Q7.21.2 8
Q7.20.1 D	Q7.18.1 2.90 to 3.10	Q7.17.1 31
Q7.17.2 2.1	Q7.16.1 6	Q7.16.2 4096
Q7.16.3 8	Q7.15.1 11	Q7.15.2 C
Q7.15.3 2	Q7.15.4 0.5	Q7.14.1 10
Q7.14.2 B	Q7.14.3 3.375	Q7.13.1 A
Q7.11.1 B	Q7.10.1 D	Q7.09.1 D
Q7.08.1 A	Q7.07.1 B	Q7.05.1 A
Q7.05.2 C	Q7.05.3 D	

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